Cloudra: Geometric Reasoning and Scientific Validation for Autonomous Reaction Mechanism Exploration

Chengyi Yang [1,*], Qianxin Zhou [2,†], and Yang Li [1,†]

[1]College of Chemistry and Pharmaceutical Engineering, Changsha University of Science and Technology, Changsha 410114, China
[2]State Key Laboratory of Photocatalysis on Energy and Environment, College of Chemistry, Fuzhou University, Fuzhou 350002, China

[*]Corresponding author: Chengyi Yang. E-mail: 202509240114@csust.edu.cn.
[†]Qianxin Zhou and Yang Li contributed equally as second authors.

2 November 2026

Abstract

Autonomous reaction-mechanism exploration requires proposing reaction steps, preparing three-dimensional structures, and determining which reactant and product a computed saddle point connects. Large language model (LLM) agents can misjudge spatial relationships and confuse a computed structure with the species or step they intended to study.

We present Cloudra, a multi-agent framework that constrains mechanistic output through a Delivery DAG of chemical objects. Species nodes carry geometries and bonding topology; elementary-step edges carry transition states and reaction connections between species. Cheminformatics algorithms establish molecular identities and atom correspondences. Deterministic routines measure and transform structures to prepare candidate geometries. Frequency and IRC calculations determine the stationary points found and the species they connect. Agents use these objects to test proposed steps, compare pathways, and revise the mechanism.

CloudraBench evaluates the framework's end-to-end mechanism-calculation capability across 50 tasks covering elementary reactions, multi-step mechanisms, stereochemical competition, and computational method limitations. Across 200 runs under four workflow configurations, the primary Guided workflow achieves a mean composite score of 0.805, compared with 0.584 for the same-model Standard baseline. On Hard-tier tasks, the Guided mean increases from 0.689 under Standard to 0.958. Under Guided, the self-consistency component reaches its maximum score in 24 of the 40 non-Impossible tasks, compared with seven under Standard. Behavioral analyses examine how the agents use bond and dihedral measurements, optimized structures, and IRC endpoints to revise local mechanistic assignments.

1 Introduction

1.1 Physical requirements for autonomous reaction mechanism research

Reaction-mechanism validation verifies that molecular geometries, reaction pathways, and quantum-chemical results correspond to the declared species and elementary steps. Frequency analysis identifies minima and first-order saddles, and intrinsic reaction coordinate (IRC) calculations identify the species connected by each saddle. The computed endpoints determine the saddle's mechanistic assignment.[1, 2]

Automated reaction-network exploration organizes calculations through chemical objects. Within SCINE, Chemoton 2.0 groups structures into compounds and elementary steps into reactions, and validates optimized saddles through IRC calculations, molecular-graph comparisons, and product optimization.[3] AutoRXN combines autonomous reaction exploration with density-functional calculations, coupled-cluster energy refinement, and automated multireference diagnostics.[4]

Computational chemistry agents automate several parts of this work. El Agente Q [5], Aitomia [6], and Gráfico [7] orchestrate quantum-chemical calculations and typed workflows. Estructural [8] and ARCHE [9] provide tools for molecular fragment editing, atom mapping, and transition-state geometry preparation; Quntur [10] addresses reaction kinetics and transition-state research. Extending these capabilities to autonomous mechanism exploration requires reliable geometric operations and physical interpretation of the species and pathways obtained.

Aligning stereochemical orientations, assembling reactant fragments, and preparing pathway endpoints require three-dimensional geometric reasoning. Language models can make errors when they infer spatial relationships or directly manipulate Cartesian coordinates. Quntur [10], for example, reports the need for human assistance with endpoint alignment in complex pathway and transition-state preparation. In Cloudra, an agent can ask the program to rotate the atoms on one side of a bond, measure the resulting dihedral angle, and compare the new geometry with another transition-state guess.

1.2 Cloudra: autonomous mechanism exploration through chemical objects

Tool-oriented workflow graphs use calculation calls as nodes and execution dependencies as edges. Cloudra is a multi-agent system for small-molecule mechanism exploration that represents the delivered reaction mechanism through species and elementary-step objects in a Delivery DAG. Agents express candidate species and elementary steps in this graph, choose geometric operations and subsequent calculations from the associated structures, energies, and pathway results, and deliver the resulting mechanism through the same objects.

Figure 1 summarizes a structured literature sample of 85 chemical AI agent systems: 39 concern materials or surfaces, 23 concern finite molecules or complexes, 12 concern soft matter or biomolecules, and 11 cover multiple categories of objects. Figures 2 and 3 show the documented tasks in the finite-molecule/complex category and compare Cloudra's capabilities with those of the existing systems. The accompanying literature materials contain the complete matrix and the sources for each entry.

Cloudra uses deterministic routines to calculate and modify molecular coordinates. MCS matching selects molecular fragments; atom mapping matches reactant and product atoms and identifies atom pairs whose bonds change. The program measures bond lengths, bond angles, dihedral angles, and stereochemistry, and rotates or translates selected atoms to prepare candidate structures.

The Delivery DAG represents the mechanism as a graph of chemical objects. Each species node carries molecular

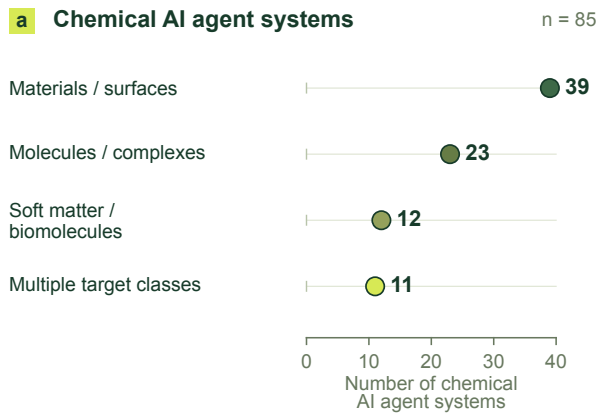


Figure 1. Physical targets in a structured literature sample of 85 chemical AI agent systems: 39 materials/surface, 23 finite-molecule/complex, 12 soft-matter/biomolecular, and 11 cross-class systems.

d Molecules / complexes n = 23 · 1/2

	AI agent system	T	E	O	F	R	D
01	Buehler molecular DFT	●	●	○	○	○	○
02	El Agente Q	●	●	●	●	○	○
03	Aitomia	●	●	●	●	○	○
04	MatAgent (Lv et al.)	●	●	○	○	○	○
05	ChemGraph	●	●	●	●	○	○
06	AutoSolvate-Agent	●	●	●	●	○	○
07	ChemAgents (azobenzene)	○	○	○	●	○	●
08	El Agente Cuántico	○	●	○	○	○	○
09	ChatMat	●	●	○	○	○	○
10	El Agente Estructural	●	○	●	○	○	○
11	El Agente Quntur	●	●	●	●	●	○
12	Chelatron	●	●	○	○	○	○
13	OpenClaw comp. chemistry	●	○	●	○	○	●
14	Protomia	●	●	○	●	○	○

● Demonstrated ○ Unencoded

Figure 2. Documented tasks in the finite-molecule/complex class (panel 1 of 2; 23 chemical AI agent systems in total). T: structure construction/editing; E: static energy/properties; O: geometry optimization; F: vibrations/thermochemistry; R: explicit pathways/TS; D: molecular/quantum dynamics. Filled cells denote positive evidence; blanks leave capability coverage unspecified.

d Molecules / complexes n = 23 · 2/2

	AI agent system	T	E	O	F	R	D
15	El Agente Forjador	●	●	●	●	○	●
16	VirtualLab_CC	●	●	●	●	○	○
17	LADeQ	○	●	○	○	○	○
18	Evolving TS workflows	○	○	●	●	●	○
19	ARCHE	●	●	●	●	●	○
20	Multireference QC agent	○	●	○	○	○	○
21	SQD-Agent	○	●	○	○	○	○
22	QAOS / Quantum Agentic Operating System	●	●	●	○	○	○
23	PULSES	●	●	●	○	○	●
	Cloudra (this work)	●	●	●	●	●	○

● Demonstrated ○ Unencoded

Figure 3. Chemical AI agent systems in the finite-molecule/complex class, continued (panel 2 of 2), using the codes in Figure 2. Cloudra (this work) is benchmarked at the bottom. Complete matrices for all 85 chemical AI agent systems and source-level qualifications accompany the draft.

geometries, bonding topology, stereochemistry, and associated calculation results. Each elementary-step edge carries the transition state and the reaction path connecting adjacent species. Topology and stereochemistry comparisons identify species, while atom mapping tracks corresponding atoms across reactant and product structures. Frequency calculations characterize the stationary points, and IRC endpoint calculations determine their reaction connections. The program checks that the frequency calculation and both IRC directions refer to the same first-order saddle, and that optimizations from the IRC terminal frames reach the declared reactant and product species. The Supervisor reviews these objects against the task and identifies calculations still needed and steps requiring further investigation.

1.3 Contributions

Cloudra's central design connects LLM research decisions and mechanism delivery through chemical objects. The Delivery DAG specifies the species and steps under investigation and the calculation results required to support them. Molecular topology and stereochemistry identify species; atom mappings track atoms across structures. Geometry operations construct candidates, and frequency and IRC calculations assign the resulting stationary points and reaction connections. The agents use these objects to extend and revise the mechanism as the study progresses.

The geometric procedures include MCS-based fragment selection, graph traversal to select the atoms that move, and transition-state fragment assembly. They allow the agents to rotate selected groups, adjust fragment positions, measure the resulting structures, and prepare new guesses after an unsuccessful transition-state search.

We evaluate the framework's end-to-end mechanism-calculation capability using CloudraBench, which measures graph coverage, physical self-consistency, and task-defined target accuracy from native computational outputs. Its 50 tasks cover elementary and multistep reactions, stereochem-

istry, and the applicability limits of the allowed methods. The experiments comprise 48 independent runs with three repeats on four tasks and a completed study of 200 runs under four workflow configurations across all 50 tasks. Standard uses the same model backbone as the primary comparison; Record and Brief-errors examine evidence organization and diagnostic feedback.

2 System Design and Methods

2.1 System Overview and Division of Responsibilities

Cloudra investigates reaction mechanisms by constructing candidate molecular configurations, locating and characterizing stationary points, tracing elementary reaction paths, and comparing competing channels under the specified reaction conditions and computational requirements. A workflow graph organizes tool calls and computational tasks by their execution dependencies. Cloudra's Delivery DAG represents the mechanism being delivered: its nodes are species occurrences, and its edges are elementary reaction steps. Agents must express candidate species and their reaction connections in this chemical graph. They choose subsequent geometric operations, calculations, and exploration branches from their hypotheses and returned results.

Canonical topology and stereochemical comparisons identify the species represented by a structure, while atom mapping identifies corresponding atoms across reactant and product configurations. Geometric operators rotate, translate, and assemble molecular fragments. Frequency calculations determine stationary-point curvature, and IRC calculations followed by endpoint optimization and frequency analysis determine which species a saddle point connects.

The Coordinator proposes mechanistic hypotheses and candidate elementary steps, and selects which branches to investigate. Computational agents prepare three-dimensional structures with the geometric operators and submit quantum-chemical calculations for their assigned steps. If a geometry diverges, a saddle-point optimization fails to converge, or a path bifurcates, the computational agent examines the structures and calculation diagnostics to decide whether to continue the step. If the step appears blocked, it reports the diagnosis to the Coordinator, which revises the hypothesis or explores another branch. The Supervisor reviews the species, steps, and calculation results in the DAG against the task requirements. Its review notes are stored separately from the facts parsed from calculation outputs.

2.1.1 Asynchronous Computation Scheduling and Resource Management

The calculation database (`ComputeDB`) stores each job's status. Separate capacity pools limit concurrent calculations on local workers, remote SSH clusters, and Slurm queues. Before dispatch, the scheduler computes a SHA-256 digest from the Cartesian coordinates, spin, charge, and level of theory. The scheduler reuses a matching active job or cached result. Operating-system exclusive file locks (`flock`) track running processes and concurrency; quantum-chemistry processes holding active locks remain running during monitor exceptions. When a calculation ends, a scheduler notification triggers asynchronous output reading by the computational agent.

2.2 Molecular Representations and Geometric Operators

A molecular input consists of an attributed graph $H = (A, B)$ and Cartesian coordinates $\mathbf{X} \in \mathbb{R}^{n\times 3}$. Here A is the atom set, B is the bond set, and each row of $\mathbf{X}$ gives one atom position. The graph stores atom and bond attributes. Geometric operators change the coordinates and return measurements of the resulting structure. Replacing atoms or functional groups also changes H.

2.2.1 Atom Correspondence and Reaction Centers

Atom attributes retain element, isotope, formal charge, hydrogen count, radical-electron count, and declared stereochemistry; bond attributes retain order and stereochemical labels. Existing coordinates are retained. ETKDGv3 [11–13] embeds the remaining components, which are then assembled if necessary. Complete, unique atom-map labels on both endpoints define the correspondence directly and must agree in element and isotope. Otherwise, let A_{P} and A_{R} be the product and reactant heavy-atom sets. Candidate bijections $\pi : A_{\mathrm{P}} \to A_{\mathrm{R}}$ satisfy $z_i^{\mathrm{P}} = z_{\pi(i)}^{\mathrm{R}}$ and $\iota_i^{\mathrm{P}} = \iota_{\pi(i)}^{\mathrm{R}}$, where z is atomic number and ι is isotope label. Each candidate is scored by

$$J(\pi) = 2(N_{\mathrm{f}} + N_{\mathrm{b}}) + N_{\mathrm{o}} + 4\Delta_H + 8\Delta_q + 8\Delta_r + 2N_{\mathrm{s}}. \tag{2.1}$$

Let B_{P}^{π} denote product bonds expressed in reactant atom indices. Bond formation and cleavage are the set differences $B_{\mathrm{P}}^{\pi} \setminus B_{\mathrm{R}}$ and $B_{\mathrm{R}} \setminus B_{\mathrm{P}}^{\pi}$, with cardinalities N_{f} and N_{b}. The count N_{o} includes pairs in $B_{\mathrm{P}}^{\pi} \cap B_{\mathrm{R}}$ whose bond orders differ. For $a \in \{H, q, r\}$, the mismatch is $\Delta_a = \sum_{i\in A_{\mathrm{P}}} |a_i^{\mathrm{P}} - a_{\pi(i)}^{\mathrm{R}}|$, where H, q, and r are the atom hydrogen count, formal charge, and radical-electron count. N_{s} counts stereochemical mismatches. Exactly one declared tetrahedral label contributes one mismatch, increasing J by 2. When both corresponding atoms carry tetrahedral labels, the comparison requires four corresponding neighbors, including a single hydrogen where applicable; it corrects for neighbor-order parity before comparing the labels. For two-sided labels, only comparable environments enter mismatch scoring.

For element–isotope classes of sizes n_g, the search space contains $N_\Pi = \prod_g n_g!$ bijections. All are enumerated when $N_\Pi \le 10^5$. Above this limit, MCS seeds use element matching, unrestricted bond-order matching, ring-to-ring matching, and complete rings [13]. The MCS timeout is 10 s, and each endpoint contributes at most 64 substructure matches. Unmatched atoms are paired within element–isotope classes after sorting by hydrogen count, degree, and input index. Candidates with the same formed, broken, and order-changed bonds and the same score are merged, keeping the lexicographically smallest mapping for each group. Representatives are sorted by score and then atom-map sequence; the first 16 form the heavy-atom candidate set C. For nonempty C, define optimal score $J_{\min} = \min_{\pi\in C} J(\pi)$ and tolerance set $\mathcal{N} = \{\pi \in C : J(\pi) \le J_{\min} + 2\}$. An unlabeled mapping is reported as unique when $|\mathcal{N}| = 1$, ambiguous when $|\mathcal{N}| > 1$, and inconclusive when $C = \emptyset$.

Each retained heavy-atom mapping is expanded to all atoms using matches compatible with both endpoint graphs. Compatible explicit hydrogen labels are preserved first; re-

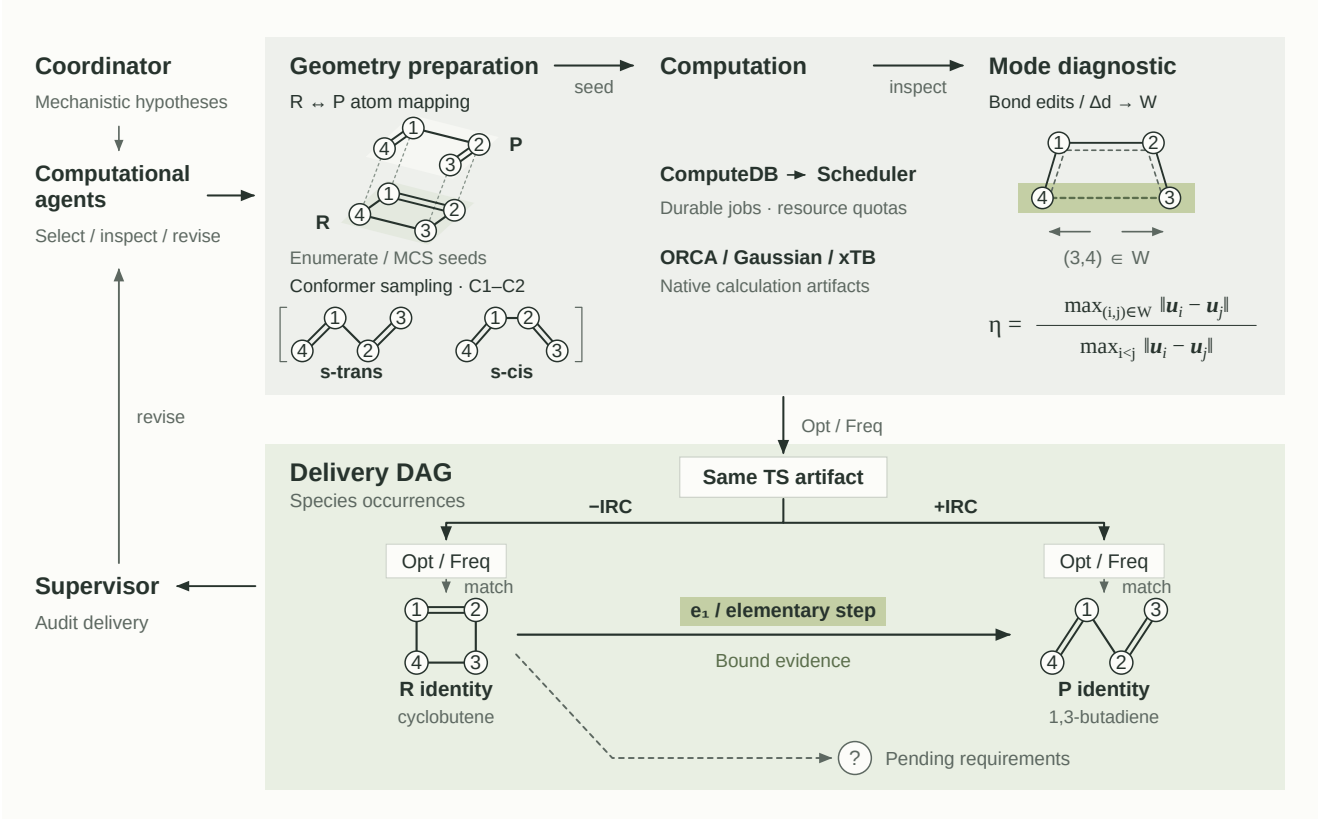


Figure 4. Cloudra's mechanism exploration workflow. Atom mapping and conformer sampling prepare reactant and product structures; ComputeDB and the scheduler manage the calculations. Mode diagnostics measure motion on monitored atom pairs. Bidirectional IRCs start from the same transition state, and their outer frames undergo endpoint optimization and frequency analysis. The resulting structures are compared with the species declared in the Delivery DAG. The Supervisor reviews the graph and identifies calculations to complete and assignments to resolve.

maining hydrogens are paired by isotope on corresponding heavy atoms. For remaining isotope groups of sizes h_g, the remaining $N_H = \prod_g h_g!$ permutations are enumerated when $N_H < 257$. An expansion with $N_H \geq 257$ requires compatible explicit hydrogen labels that resolve the permutation space; unresolved expansions are reported as inconclusive. Complete maps are deduplicated, ranked by score and atom-map sequence, and evaluated with the same +2 ambiguity criterion. The selected all-atom bijection reorders the product into reactant atom order. Formed and broken bonds identify the atom pairs monitored during path interpolation; bond-order changes are recorded with the transformed graph.

2.2.2 Structural Measurements and Rigid Body Operations

The system calculates bond lengths, bond angles, dihedral angles, ring conformations, and close contacts from the coordinates. For coordinates $\mathbf{x}_i$, the interatomic distance is $d_{ij} = \|\mathbf{x}_j - \mathbf{x}_i\|$; the bond angle at vertex j is obtained from the normalized dot product of the two bond vectors. Distances are compared with reference bond lengths and van der Waals contact scales to flag geometric anomalies. For coordinate-only inputs, connectivity is determined at entry. Local coordinate transformations retain this bond graph and repeat the measurements before and after moving atoms.

Default fragment selection on either side of an acyclic bond uses graph connectivity: the specified bond $a-b$ is temporarily removed, and the connected component containing b, denoted $\mathcal{F}_b$, is selected. If the endpoints remain connected after bond removal, movement within a ring framework requires an explicit atom group. Let $\mathcal{F}$ denote the explicitly specified moving atom group or the component $\mathcal{F}_b$ obtained by bond removal. To rotate the selected fragment about an axis, Rodrigues' formula is applied to every $i \in \mathcal{F}$:

$$\begin{aligned} \mathbf{x}_i' &= \mathbf{a} + \mathbf{R}(\hat{\mathbf{u}}, \theta)(\mathbf{x}_i - \mathbf{a}), \\ \mathbf{R} &= \cos\theta\, \mathbf{I} + (1 - \cos\theta)\hat{\mathbf{u}}\hat{\mathbf{u}}^{\mathsf{T}} \\ &\quad + \sin\theta\, [\hat{\mathbf{u}}]_{\times}. \end{aligned} \tag{2.2}$$

Here, $\mathbf{a}$ is a point on the axis, $\hat{\mathbf{u}}$ its unit direction, θ the rotation angle, $\mathbf{I}$ the identity matrix, and $[\hat{\mathbf{u}}]_{\times}\mathbf{v} = \hat{\mathbf{u}} \times \mathbf{v}$. The operation fixes the coordinates of atoms outside the selected set. Stretching bond $a - b$ translates $\mathcal{F}_b$ as a whole:

$$\mathbf{x}_i' = \mathbf{x}_i + (L - d_{ab})\frac{\mathbf{x}_b - \mathbf{x}_a}{d_{ab}}, \qquad i \in \mathcal{F}_b. \tag{2.3}$$

Here L is the target bond length; coordinates outside $\mathcal{F}_b$ remain fixed. Chirality inversion rotates two separable branches around their center to exchange branch directions. Group substitution removes the specified-side fragment, docks the new group along the original attachment-bond direction, and updates the graph. To compare structures before and after optimization, the system differences distance-perceived connectivity sets under the same element sequence and classifies gained or lost contacts as intra- or interfragment changes.

2.2.3 Branch Crowding and Reactant Preorganization

After checking hybridization, ring structure, and substituent graph inequivalence, the system calculates crowding around each candidate axis from the atoms in its substituent branches:

$$B_s = \sum_{k \in T_s} c_k r_k \rho^{\ell_k}, \qquad \rho = 0.5, \quad 0 \le \ell_k \le 5. \tag{2.4}$$

Here r_k represents the van der Waals radius of atom k. T_s is generated by a last-in-first-out depth-first traversal starting from the substituent root at depth zero, with the adjacent axis-end atom marked visited. On popping an unvisited atom, the algorithm records its traversal depth ℓ_k and pushes its unvisited neighbors if $\ell_k < 5$. The weighting coefficient is $c_k = 1.5$ for a root of graph degree at least 3, and $c_k = 1$ otherwise.

The descriptor uses standard van der Waals radii in Å: H 1.20, B 1.92, C 1.70, N 1.55, O 1.52, F 1.47, P 1.80, S 1.80, Cl 1.75, Br 1.85, and I 1.98 (1.70 fallback for unlisted elements). The four branch values are rounded to three decimal places and summed to obtain the axis score B_{axis}; $B_{\text{axis}} \ge 9.0$ flags the axis as congested for subsequent conformational inspection.

During multicomponent preorganization, the primary fragment is fixed at the origin. Remaining fragments undergo orientation sampling on the unit sphere. With K samples, direction s is

$$\begin{aligned} \hat{\mathbf{u}}_s &= (a_s \cos s\gamma,\ y_s,\ a_s \sin s\gamma), \\ a_s &= \sqrt{1 - y_s^2}, \\ y_s &= 1 - \frac{2s}{K-1}, \\ \gamma &= \pi(3 - \sqrt{5}). \end{aligned} \tag{2.5}$$

Here $\gamma = \pi(3 - \sqrt{5}) \approx 2.3999$ rad $(137.5°)$ represents the Golden Angle, generating a Fibonacci spherical spiral grid for uniform distribution of directions. The axial twist about $\hat{\mathbf{u}}_s$ is $2\pi s/K$ (default $K = 32$). Translation along the interfragment vector resolves steric overlap. For a set W of guided atom pairs with target distance windows $[l_{ij}, u_{ij}]$, placement $\mathbf{X}$ is scored by

$$S(\mathbf{X}) = \max_{(i,j) \in W} \{0,\ l_{ij} - d_{ij}(\mathbf{X}),\ d_{ij}(\mathbf{X}) - u_{ij}\}. \tag{2.6}$$

Configurations with low $S(\mathbf{X})$ undergo distance-restrained MMFF94 force-field optimization[14] to bring the specified atom pairs into their target distance windows while relaxing nonbonded contacts. Candidates are retained only if the force-field optimization converges, the distances fall within their windows, stereochemistry is preserved, and fragment contacts satisfy $d_{i^* j^*} \le 1.7(r_{i^*} + r_{j^*})$. The retained configurations are ranked by unconstrained single-point energy and clustered with a heavy-atom Kabsch RMSD threshold of 0.5 Å to provide starting structures.

2.2.4 Endpoint Alignment and IDPP Interpolation

After reordering the product atoms to match the reactant, each endpoint is centered. Let $\mathbf{P}$ and $\mathbf{R}$ be the centered product and reactant coordinate matrices. Kabsch alignment[15] solves

$$\begin{aligned} &\min_{\mathbf{Q}^\mathsf{T}\mathbf{Q}=\mathbf{I}, \det \mathbf{Q}=1} \|\mathbf{P}\mathbf{Q} - \mathbf{R}\|_\mathrm{F}^2, \\ &\mathbf{P}^\mathsf{T}\mathbf{R} = \mathbf{U}\mathbf{\Sigma}\mathbf{V}^\mathsf{T}, \\ &\mathbf{Q} = \mathbf{U}\,\mathrm{diag}\big(1, 1, \det(\mathbf{U}\mathbf{V}^\mathsf{T})\big)\mathbf{V}^\mathsf{T}. \end{aligned} \tag{2.7}$$

The resulting matrix $\mathbf{Q}$ is the proper rigid-body rotation matrix. The determinant correction factor $\det(\mathbf{U}\mathbf{V}^\mathsf{T})$ enforces $\det \mathbf{Q} = +1$, ruling out improper reflections and preserving chiral configuration.

For M images including endpoints, define $\lambda_k = k/(M-1)$ and initialize $\mathbf{X}_k = (1-\lambda_k)\mathbf{R} + \lambda_k \mathbf{P}\mathbf{Q}$. The image-dependent pair potential (IDPP) method [16] interpolates each image's target pair distances between the endpoints and optimizes its coordinates with the potential

$$\begin{aligned} d^*_{ij,k} &= (1-\lambda_k) d^\mathrm{R}_{ij} + \lambda_k d^\mathrm{P}_{ij}, \\ V_k &= \sum_{i<j} \frac{(d_{ij,k} - d^*_{ij,k})^2}{d^4_{ij,k}}. \end{aligned} \tag{2.8}$$

The inverse-fourth-power weighting $d^{-4}_{ij,k}$ imparts stiffness to short covalent bonds, prioritizing the preservation of basic covalent distances while allowing long-range nonbonded contacts to relax. Standard mode couples intermediate images with improved-tangent nudged elastic band (NEB) springs and optimizes with MDMin, evaluating convergence by the maximum force residual across movable atoms.[17, 18] Lightweight mode evaluates target distances with an objective denominator $\max(d^*_{ij,k}, 0.1\,\text{Å})^4$ and relaxes intermediate images independently using Adam optimization.[19]

Active monitoring pairs W incorporate bond-state changes, pairs bonded at both endpoints with $|\Delta d| \ge 0.15$ Å, and the five atom pairs exhibiting the largest endpoint distance variations. Within the monitoring set W, the primary reaction coordinate and its midpoint distance are

$$\begin{aligned} p^* &\in \mathrm{argmax}_{p \in W} |d^\mathrm{P}_p - d^\mathrm{R}_p|, \\ m &= (d^\mathrm{R}_{p^*} + d^\mathrm{P}_{p^*})/2. \end{aligned}$$

The initial transition-state guess is extracted from the first image k satisfying $(d_{p^*,k} - m)(d^\mathrm{R}_{p^*} - m) \le 0$, or the image minimizing $|d_{p^*,k} - m|$. The selected configuration serves as the initial seed for saddle-point optimization.

2.3 Chemical Species and Reaction Steps in the Delivery DAG

2.3.1 Species Occurrences and Elementary Steps

The Delivery DAG is $G = (V, E)$. Each vertex $v \in V$ represents a species at one position in the reaction sequence, and each directed edge $e = (v_i, v_j) \in E$ represents an elementary step. The mapping $\kappa : V \to \mathcal{S}$ gives the canonical chemical identity of each species occurrence. An initial and a regenerated catalyst can therefore occupy separate nodes while sharing the same chemical identity.

Each species is described by molecular geometry, bonding topology, and stereochemistry. Its incoming and outgoing step edges describe its reaction connections to adjacent species. Canonical molecular graphs and stereochemical labels are compared when assigning structures to species; atom mapping identifies corresponding atoms in reactant and product structures. Species nodes record stationary-point geometries, frequencies, and absolute energies; step

edges record transition-state geometries, imaginary-mode displacement vectors, and reaction paths. Frequency and IRC calculations characterize these stationary points and paths, with the physical criteria defined below.

For each species or step, K lists the required calculation stages. Standard elementary steps require the sequence $K = \{\text{ts_opt}, \text{frequency}, \text{short_irc}\}$, with thermochemical corrections added when required by the task rubric. Calculation outputs are archived as immutable records identified by content digests. Each parsed result is recorded as $f = (c, o, r, d)$: c names the check, $o \in \{+, -, ?\}$ records support, contradiction, or uncertainty, r identifies the source calculation output, and d contains the measured values.

For a species or step t, let F_t contain its parsed results, R its required property checks, and O its optional checks. The physical validation criterion is

$$\begin{aligned} S_t = \mathbf{1}[R \neq \emptyset] \\ \wedge \bigwedge_{c \in R} \mathbf{1}[\exists f \in F_t : f.c = c,\ f.o = +] \\ \wedge \mathbf{1}[\nexists f \in F_t : f.c \in R \cup O,\ f.o = -]. \end{aligned} \tag{2.9}$$

When $R \neq \emptyset$, each required check must have a supporting result, and every current result for a check in $R \cup O$ must indicate support or uncertainty. When $R = \emptyset$, only the calculation-stage requirements are checked.

2.3.2 Curvature and Reaction Mode Diagnostics

Stationary-point curvature is evaluated from the mass-weighted Hessian:

$$\begin{aligned} \mathbf{H}_{\text{mw}} &= \mathbf{M}^{-1/2} \nabla_{\mathbf{x}}^2 U\, \mathbf{M}^{-1/2}, \\ \mathbf{H}_{\text{mw}} \mathbf{v}_k &= \lambda_k \mathbf{v}_k. \end{aligned} \tag{2.10}$$

In the vibrational subspace orthogonal to translational and rotational degrees of freedom, negative eigenvalues $\lambda_k < 0$ correspond to imaginary harmonic frequencies $\nu_k < 0$. The number of imaginary frequencies is $n_{\text{imag}} = \sum_k \mathbf{1}[\nu_k < 0]$. Ground-state reaction intermediates require $n_{\text{imag}} = 0$, whereas transition states require strictly $n_{\text{imag}} = 1$.

The geometry used for the frequency calculation is compared with the optimized stationary-point geometry. For coordinate matrices $\mathbf{X}$ and $\mathbf{Y}$ centered at the origin, the geometry match requires

$$\begin{aligned} \delta_\pi(\mathbf{X}, \mathbf{Y}) &= \max_i \|\mathbf{Y}_{\pi(i)} \mathbf{Q}_\pi - \mathbf{X}_i\| \\ &\leq 3 \times 10^{-6}\ \text{Å}, \end{aligned}$$

evaluated across element-conserving permutation mappings π and optimal Kabsch rotations $\mathbf{Q}_\pi$.

For a first-order saddle point with imaginary normal-mode displacement vector $\mathbf{u}$, the mode-alignment diagnostic compares the largest relative-displacement magnitude among monitored reaction-center pairs W with the largest among all atom pairs:

$$\eta = \frac{\max_{(i,j) \in W} \|\mathbf{u}_i - \mathbf{u}_j\|}{\max_{i<j} \|\mathbf{u}_i - \mathbf{u}_j\|}. \tag{2.11}$$

The diagnostic $\eta \in [0, 1]$ measures reaction-center participation relative to the largest pair displacement, using a comparison ratio of 0.5. It flags spectator motion such as methyl wagging or solvent libration for inspection. The alignment flag is advisory.

2.3.3 IRC Direction and Endpoint Consistency

Intrinsic reaction coordinate (IRC) trajectories evolve from the saddle point along the steepest-descent path in mass-weighted coordinates $\mathbf{q} = \mathbf{M}^{1/2}\mathbf{x}$ [1, 2]:

$$\begin{aligned} \frac{\mathrm{d}\mathbf{q}_\pm}{\mathrm{d}s} &= -\frac{\nabla_{\mathbf{q}} U}{\|\nabla_{\mathbf{q}} U\|}, \\ \mathbf{q}_{\text{init},\pm} &\approx \mathbf{q}_{\text{TS}} \pm \epsilon \mathbf{v}_-. \end{aligned} \tag{2.12}$$

Cloudra checks that both IRC directions start from the verified saddle point and extracts the terminal frame from each direction. For paths between endpoint minima, each terminal frame is used to start an unconstrained geometry optimization. The resulting stationary point must have zero imaginary frequencies ($n_{\text{imag}} = 0$), and its canonical isomeric SMILES must equal that of the declared reactant or product, including topology and stereochemistry.

For elementary steps yielding dissociated molecular components, each dissociated product must independently converge to a verified minimum. Over the monitored pair set W, let $\mathbf{q}_{\text{TS}}$, $\mathbf{q}_{\text{E}}$, and $\mathbf{q}_{\text{R}}$ denote distance vectors at the transition state, terminal IRC frame, and reactant reference. With covalent cutoffs $c_{ij} = 1.25(r_i + r_j)$ and tolerance $\tau = 0.10$ Å, define $W_{\text{c}} = \{(i,j) \in W : d_{ij}^{\text{R}} \leq c_{ij}\}$. The interval $[a_p, b_p]$ has endpoints

$$\begin{aligned} a_p &= \min(d_p^{\text{TS}}, d_p^{\text{R}}) - \tau, \\ b_p &= \max(d_p^{\text{TS}}, d_p^{\text{R}}) + \tau. \end{aligned}$$

The Boolean predicate is_between is satisfied when both conditions hold:

$$\begin{aligned} \|\mathbf{q}_{\text{E}} - \mathbf{q}_{\text{R}}\|_2 < \|\mathbf{q}_{\text{TS}} - \mathbf{q}_{\text{R}}\|_2, \\ a_p \leq d_p^{\text{E}} \leq b_p \quad \text{for all } p \in W_{\text{c}}. \end{aligned}$$

Dissociated-product paths must satisfy both the component-minimum checks and is_between.

2.3.4 Evidence Aggregation and Energies

For elementary step e, let K_e list its required calculation stages. Set $P_{e,k} = 1$ when the calculation results assigned to stage k satisfy its physical criteria, and $P_{e,k} = 0$ otherwise. All required stages must pass:

$$A_e = \bigwedge_{k \in K_e} P_{e,k}. \tag{2.13}$$

When explicit property checks R are declared, the step must also satisfy $A_e \wedge S_e$. Each stage is recorded as supported, pending, skipped, or rejected. A documented exemption or method limitation is thus recorded separately from a completed calculation.

The electronic activation barrier is computed directly from verified stationary-point energies:

$$\Delta E_e^{\ddagger} = E_{\text{TS},e} - E_{\text{ref},e}. \tag{2.14}$$

Energies share a consistent quantum chemical level of theory. For thermochemical profiles, zero-point energy, enthalpy, and Gibbs free-energy corrections are evaluated from harmonic partition functions under declared temperature and pressure specifications.

2.4 Calculation Diagnostics and Retries

The system distinguishes execution errors, numerical nonconvergence, and physical evidence problems. Execution errors include invalid input syntax, environment faults, and parsing failures. Numerical checks record self-consistent field (SCF) and geometry optimization convergence. Physical evidence problems include saddles with $n_{\text{imag}} \neq 1$, IRC endpoints awaiting verification, or molecular graphs corresponding to a different species. Imaginary-mode participation and geometric interval measurements indicate which atomic motions or distances to inspect.

Let T contain the species and elementary steps requested by the task. Extending $P_{e,k}$ to each $t \in T$, the calculation-stage requirements to be resolved are

$$U = \{(t,k) : t \in T,\, k \in K_t,\, P_{t,k} = 0\}. \tag{2.15}$$

For each $(t,k) \in U$, diagnostic feedback identifies the affected species or step and the problem: SCF nonconvergence, normal-mode misalignment, a missing IRC-to-endpoint calculation link, or a molecular graph mismatch. After a failed calculation, the system retains its output, updates U, and reports the geometry and convergence information to the agent.

Calculations that depend on a failed stage are rescheduled; earlier attempts and raw outputs are retained. The system reorders atom indices, assigns each calculation result to its species or step, and updates the graph status. The model revises the chemical constraints, dihedral angles, or electronic-structure method for the next attempt.

2.5 Implementation Defaults

The implementation described here corresponds to Cloudra revision 70daa5a.[20] Numerical arrays and linear algebra use NumPy[21]; quantum-chemical quantities are extracted with cclib alongside engine-specific native-log readers[22]; and benchmark graph representation and matching use NetworkX[23]. Preorganization uses 32 rigid orientations, up to 8 guided placements, minimum interfragment separation 1.0 Å, fragment-contact multiplier 1.7, and heavy-atom RMSD clustering threshold 0.5 Å. MMFF94 relaxation uses up to 2000 iterations per phase with a distance restraint force constant of 100.0. Standard IDPP employs 10 images optimized via MDMin (100 steps, force tolerance $f_{\max} = 0.1$, NEB spring constant $k = 0.1$); lightweight IDPP uses Adam optimization (4000 steps, step size 0.02, gradient tolerance 10^{-7}). Path convergence criteria evaluate the maximum force residual on intermediate path images.

The mapping coefficients and the $J_{\min} + 2$ ambiguity window are fixed ranking heuristics. Charge and radical mismatches carry larger costs than hydrogen-count and bond-edit mismatches. Branch crowding uses the fixed decay $\rho = 0.5$, traversal through depth five, and congestion flag $B_{\text{axis}} \geq 9.0$.

Cloudra also provides a conformer-search tool with RDKit and external CREST backends.[24] The optional CREST backend samples conformers or noncovalent complexes, followed by quantum-chemical optimization and physical validation.

2.6 CloudraBench construction and experimental design

2.6.1 Task coverage and tier rationale

CloudraBench evaluates end-to-end mechanism calculations on 50 tasks: 20 Easy tasks examine elementary transformations, conformational changes, and basic execution reliability; 10 Normal tasks examine intermediate identification and multistep continuity; 10 Hard tasks examine competing pathways and stereochemical identity; and 10 Impossible tasks examine the limits of the allowed methods and justified negative conclusions (Figure 5). Related reactions can appear in different tiers when they require different pathways, configurations, or electronic states. Tasks were selected using literature sources and reference calculations, considering both their ability to distinguish workflow performance and their computational cost. An accompanying catalogue lists each task's name, input conditions, and allowed methods.

2.6.2 Reference answers and evaluation protocol

Each task fixes the chemical conditions, charge, spin, deliverables, and allowed methods. Easy, Normal, and Hard tasks permit xTB[25], PM7[26], and B3LYP-D3(BJ)/def2-SVP[27–32]; Impossible tasks are restricted to xTB and PM7. Computational evidence is read from native ORCA, Gaussian, or MOPAC logs.[33–36] Reference answers specify species, elementary steps, stereochemical identities, and applicable numerical or negative targets. Numerical barriers are electronic energy differences at a consistent level of theory; stable-minimum references and actual IRC connectivity are verified separately. For each accessible reaction step requiring verification, we use TS frequency analysis, bidirectional IRCs, and endpoint optimization and frequency analysis. We also check that each endpoint optimization starts from the actual terminal IRC frame. Species-property and electronic-state questions are checked against their specified targets.

2.6.3 Scoring and study configurations

Graph structure, self-consistency, and accuracy are evaluated separately and combined with weights of 0.1, 0.3, and 0.6. Graded physical support and runtime delivery completeness are also reported. A correctly identified method limitation is scored as a supported negative conclusion. Each task lists reference-graph availability, available reference calculations, task-specific exceptions, and unresolved validation checks.

2.6.4 Repeated-task study, primary comparison, and workflow ablations

The repeated-task study (P1b) compares Guided, Standard, Record, and Brief-errors on E1, NB1, HA1, and IA1, with three independent runs per combination (48 task-level runs). The repeated runs used DeepSeek-v4.1-flash (1017).[39] E1, NB1, HA1, and IA1 concern elimination preorganization, multistep decarboxylation, stereochemical ring opening, and electronic-structure method limits, respectively; their archived trajectories supply the subsequent local diagnostic cases. Its elapsed-time unit is a four-task batch at concurrency two, with three batches per configuration. The full-benchmark study (P2) retains one run for each of the 50 tasks and four configurations (200 task-level runs). All P2 configurations used DeepSeek-v4.1-flash (1017), called with the

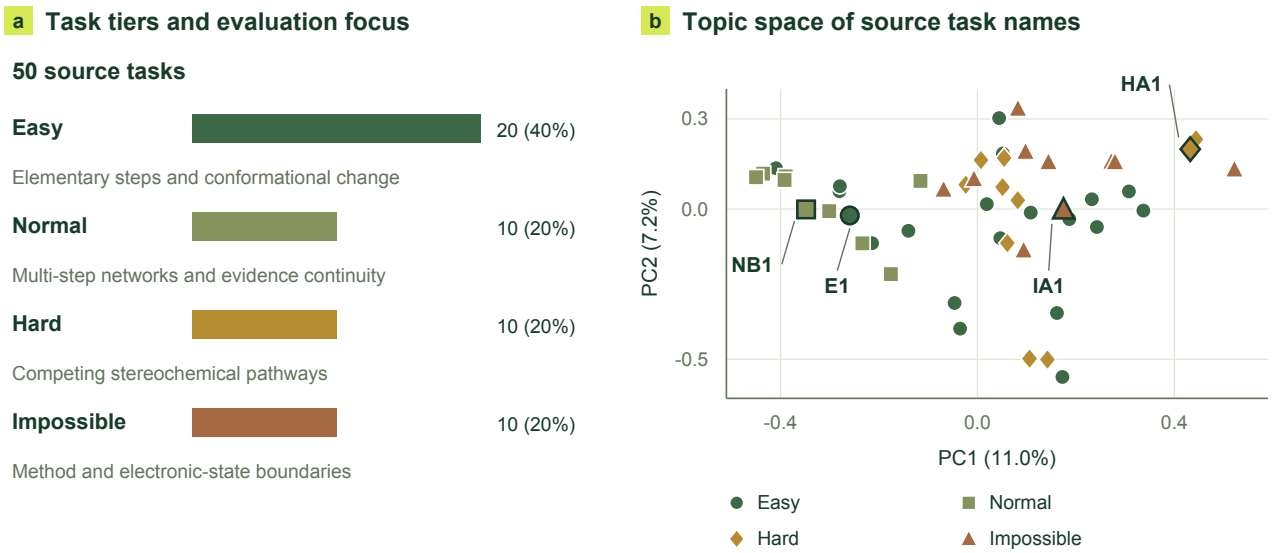


Figure 5. CloudraBench task composition and research themes. (a) Counts and evaluation focus of the four task tiers. (b) Two-dimensional PCA, computed with scikit-learn[37], of BAAI/bge-m3 embeddings[38] of all 50 original task names. E1, NB1, HA1, and IA1 are the four tasks used in the repeated-run experiment. Colors and marker shapes encode the predefined task tiers.

archived API alias `deepseek-v4-flash` through the official DeepSeek endpoint and streaming OpenAI-compatible chat completions, with requested minimal reasoning effort. The primary comparison is Guided against Standard using this same backbone; Standard is labelled Backbone in the figures. The P2 execution versions and request settings are listed in Supplementary Information Section S1.

Standard hides explicit DAG tools, removes enforcement of physical checks, and supplies neutral computational feedback. Guided uses the research graph, scientific checks, and detailed diagnostics. Record provides tools and records with optional physical validation; Brief-errors retains guidance while shortening diagnostics. All configurations receive the same task statements and method restrictions. The archive stores each run's code version, budget and stopping reason, together with the source files for its calculations.

Before scoring, human curators reconstruct each run's mechanism graph from its conversations, tool records, reports, and calculation files, following the agent's stated mechanistic intent and conclusions. They restore file references and record the source of each recovered field. Reconstruction uses only that run's existing calculations and conclusions. The evaluator searches its sandbox for native calculation evidence and scores the graph against Gold targets under common rules. The final dataset contains all 200 runs and reports Graph, Self-consistency, and Accuracy alongside each composite score. Paired differences and 95% percentile intervals use 100,000 task-cluster bootstrap samples, retaining the four configurations of each task together (seed 20261031). Computational effort is measured on the 15 selected tasks with records for all four configurations, using quantum-chemical job counts and cumulative job-hours.

3 Results and Discussion

3.1 A two-step mechanism expressed through chemical objects

The Guided NI1 run computes methylamine addition to formaldehyde and carbinolamine dehydration to an imine–water complex (Figure 6). Its Delivery DAG connects the optimized reactant complex, carbinolamine, and product complex through two first-order saddles. Calculations used B3LYP-D3(BJ)/def2-SVP in the gas phase, with charge zero and singlet multiplicity.

The addition saddle has one imaginary frequency at -1481.06 cm^{-1} and an electronic barrier of 28.12 kcal/mol relative to the reactant complex. Its two short IRC branches lead, after endpoint optimization and frequency analysis, to the reactant complex and carbinolamine, both with zero imaginary frequencies. The dehydration saddle has one imaginary frequency at -1796.84 cm^{-1} and an electronic barrier of 57.21 kcal/mol relative to its IRC-side carbinolamine minimum. Its two branches similarly yield carbinolamine and the imine–water complex. Each IRC followed 24 points in each direction before reaching the point limit; the endpoint calculations establish the resulting minima.

Interconversion between the carbinolamine conformers remains uncomputed. Supplementary Information Section S4 provides the conformer references, absolute energies, native calculation records, and figure inputs.

3.2 End-to-end benchmark evaluation

CloudraBench measures end-to-end mechanism-calculation capability: constructing candidate mechanisms, carrying out the required quantum-chemical calculations, and reaching task-consistent positive or negative conclusions. We report composite scores together with their graph, self-consistency, and accuracy components. Offline postprocessing reconstructs mechanism graphs from each run's own research materials and compares the computed species, reaction channels, electronic states, and barrier references with the task definition. Before calculating Accuracy, we match each reported reaction channel and electronic state to the target specified in the task. All four workflows use the same postprocessing rules. The resulting 200-run table contains all three components for every run, with an overall composite mean of 0.7176. Figures 7–11 report runtime graph production, TS–IRC–endpoint evidence, and agreement with the task's answer.

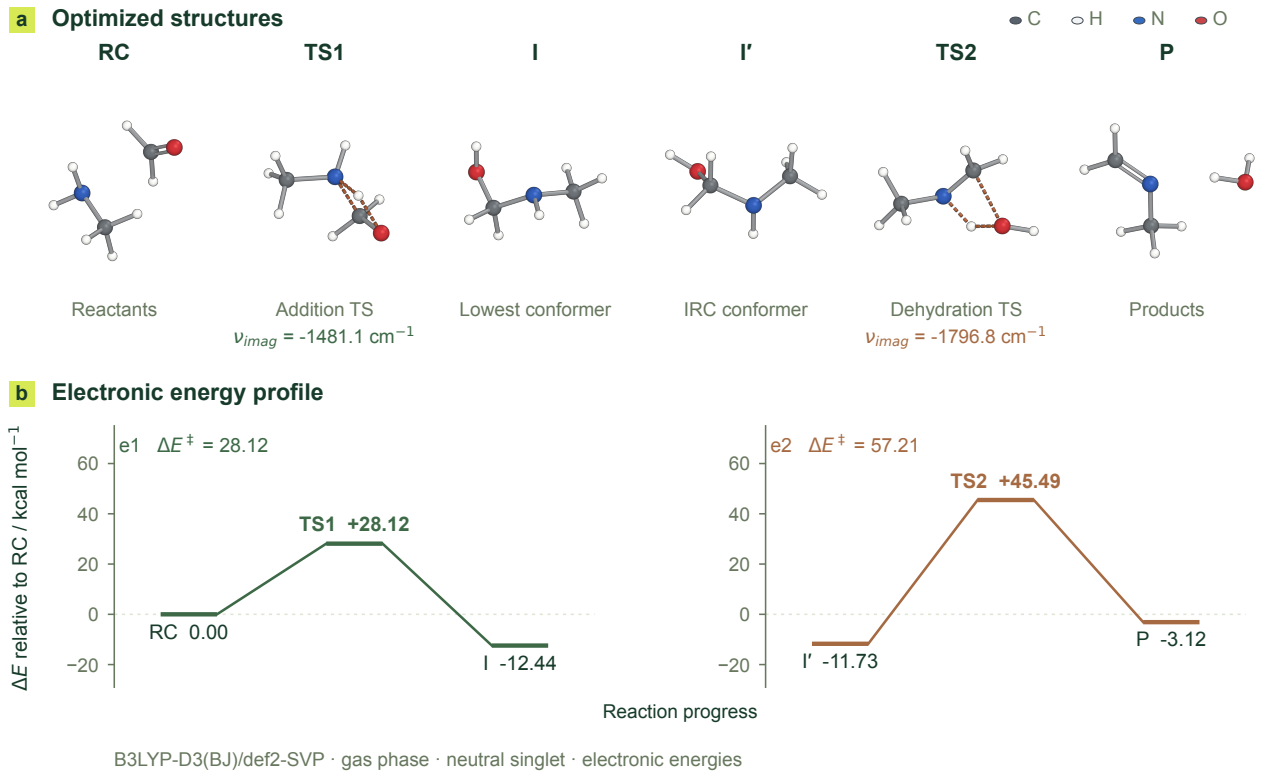


Figure 6. Two requested steps in the archived Guided/NI1 study: methylamine addition to formaldehyde and carbinolamine dehydration. Structures and electronic energies come from this run's B3LYP-D3(BJ)/def2-SVP gas-phase calculations. Each saddle has one imaginary frequency, and its short IRC branches lead through endpoint optimization/frequency analysis to minima with zero imaginary frequencies. Both energy panels use RC as the common reference.

3.2.1 Cloudra improves end-to-end mechanism-calculation scores with the same backbone

Guided obtained a mean composite score of 0.8050, compared with 0.5836 for Standard. Across the 50 task pairs, the mean improvement was 0.2214, with a 95% task-bootstrap interval of [0.1349, 0.3124]. Guided scored higher on 35 tasks, lower on eight, and equally on seven; the median paired difference was 0.1142 (Figure 7).

Guided's mean over its ten lowest-scoring tasks was 0.3374, compared with 0.0930 for Standard.

The most consistent difference occurred in the Hard stereochemistry tasks. Guided averaged 0.9577 and Standard 0.6888, a paired increase of 0.2689 [0.1213, 0.4459], with Guided scoring higher on all ten tasks. These tasks require assigning each barrier and computed pathway to the corresponding stereoisomer or competing channel.

On Normal tasks, Guided averaged 0.7370 versus 0.5731, an increase of 0.1640 [0.0365, 0.2922], with seven wins and three losses. Easy-task means were 0.9125 and 0.7044, respectively; the paired median difference was 0.0209, showing a different pattern from the broadly positive Hard comparison. Impossible-task means were 0.5051 and 0.2473. Their interpretation depends on whether the run addressed the requested method-limited negative target, so these tasks are examined separately below.

3.2.2 Self-consistency contributes most to the composite-score gain

The three components score different parts of the calculation (Figure 8). Graph measures the delivered coverage and identities of the requested species and steps. Self-consistency measures the support linking saddle points, frequencies, IRC starts, and endpoint optimization/frequency calculations. Accuracy measures agreement with the task's numerical or negative target.

Across all 50 Guided–Standard task pairs, the weighted graph contribution to the mean improvement was +0.0108, self-consistency contributed +0.1179, and accuracy +0.0927, accounting together for the composite gain of 0.2214. For the 20 Normal and Hard tasks, the mean gain was 0.2164, with contributions of +0.0080 from graph, +0.1247 from self-consistency, and +0.0837 from accuracy. Within the ten Normal tasks, the corresponding contributions were +0.0098, +0.1141, and +0.0400; within the ten Hard tasks they were +0.0063, +0.1353, and +0.1274.

On Hard tasks, the weighted accuracy improvement is comparable in size to the self-consistency improvement.

HH1 provides a direct example. Guided and Standard both reported a binaphthyl racemization barrier of 23.48 kcal/mol, close to the 23.01 kcal/mol reference. Both had graph scores of 1 and accuracy scores of approximately 0.973. Their self-consistency scores were 1.0 and 0.5, producing composite scores of 0.9838 and 0.8338, respectively. The archived Standard study contained candidate channels and IRC inputs; the required endpoint connections remained unverified. Guided attached the path evidence to the corresponding saddle and endpoints. HI1 shows a related pattern: accuracy was approximately 1 in both runs, while self-consistency increased from 0.2333 to 1.0 and the composite score from 0.7032 to 0.9999.

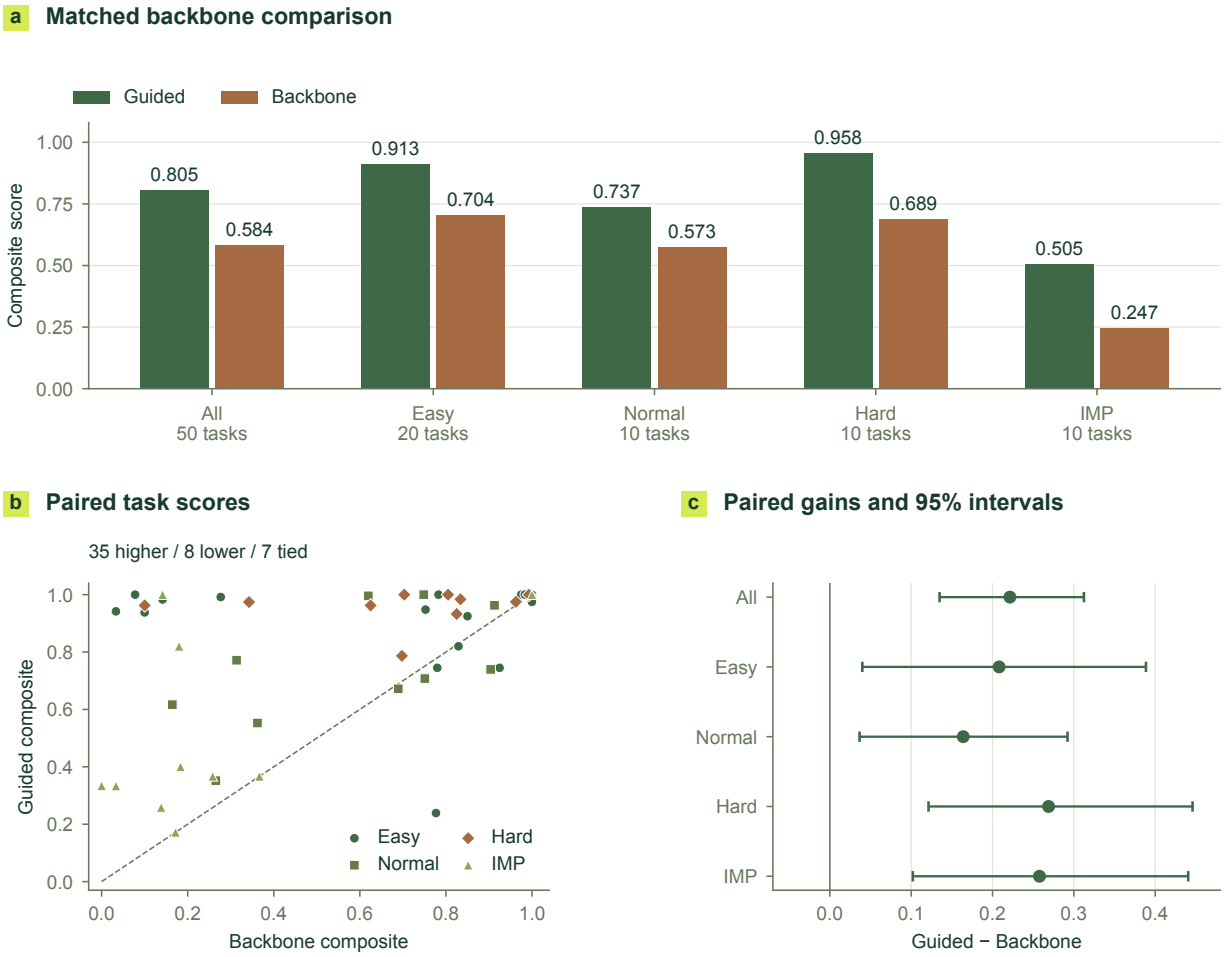


Figure 7. P2 benchmark outcomes under the archive-adjudicated track. (a) Guided and the same-backbone Standard baseline across all 50 tasks and each tier. (b) All 50 paired task outcomes. (c) Mean paired score differences with 95% percentile intervals from 100,000 task-level bootstrap resamples. Each task has one run per configuration; Easy contains 20 tasks and the other tiers contain 10 each.

3.2.3 Mechanism-graph delivery and self-consistency scores

Runtime root-graph production is recorded separately from the physical evidence found in each run (Figure 9). During the original runs, Standard delivered root graphs for 43 of 50 tasks, while Guided, Record, and Brief-errors each delivered 50. The seven Standard tasks requiring offline root-graph reconstruction were F1, HA1, HF1, IA9, M17, NR1, and P3. Offline organization assembled their graphs from the corresponding runs' existing materials, producing a complete 50-task set for every configuration.

The seven Standard studies averaged 0.1537 after organization, compared with 0.8285 for Guided on the same tasks, and contributed 42.7% of the total Guided–Standard difference. Their individual scores ranged from 0.0333 to 0.3425.

For the 40 non-Impossible tasks in each configuration, Guided had self-consistency exactly equal to 1 on 24 tasks (60.0%) and partial positive credit on 16. Standard had seven full-score (17.5%), 29 partial-score, and four zero-score tasks. Record had 16 full-score (40.0%) and 24 partial-score tasks; Brief-errors had 20 full-score (50.0%) and 20 partial-score tasks. Figure 9 presents these complete evidence-score distributions alongside the separate runtime graph-delivery counts.

These percentages count full self-consistency scores for delivered species and steps. A uniform audit covering every species and step requested by each task remains pending (Supplementary Information Section S4).

3.2.4 Guided outperforms Record with the same available tools

Record averaged 0.7038 and Brief-errors 0.7781 across all 50 tasks. Guided exceeded Record by 0.1012 [0.0396, 0.1695], with 28 wins, seven losses, and 15 ties. On all 20 Normal and Hard tasks together, the mean Guided–Record difference was 0.1468: graph contributed +0.0083, self-consistency +0.0919, and accuracy +0.0466. The largest contribution came from self-consistency, which scores the frequency, IRC, and endpoint calculations supporting the reported steps.

3.2.5 Research batch times and quantum-chemical job-hours

Figure 10 presents the timing information from the repeated-task study alongside the computational effort available for P2. For the repeated study, the measurement unit is a complete four-task batch at a concurrency of two, with three observed batches per configuration. Mean batch elapsed times were 4.451, 3.958, 3.894, and 4.827 hours for Guided, Standard, Record, and Brief-errors, respectively. Guided batches took 12.5% longer than Standard and 7.8% less time

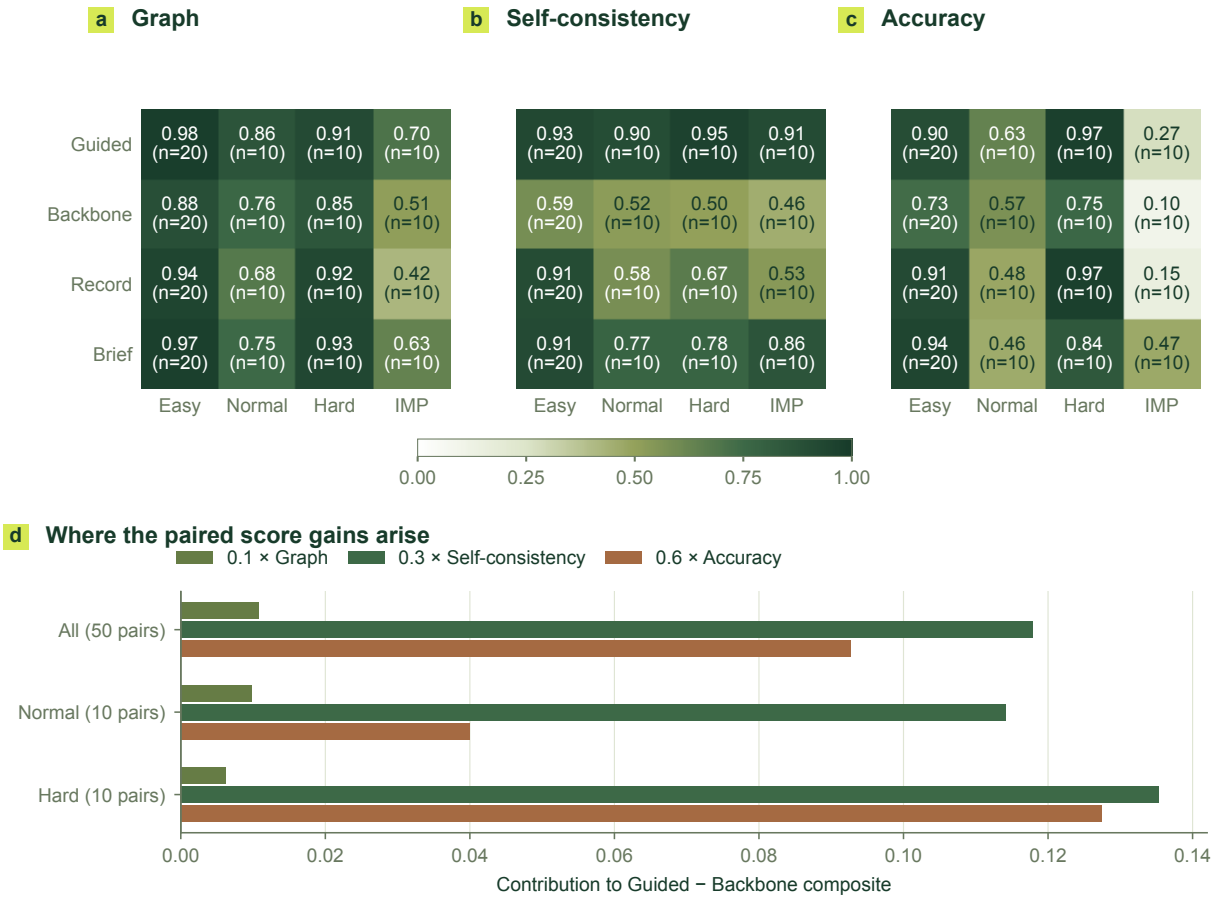


Figure 8. CloudraBench score layers across all 200 organized runs. (a–c) Means of Graph, Self-consistency, and Accuracy in each arm and tier, using 50 tasks per arm overall, 20 Easy tasks, and ten tasks in each other tier. (d) Guided–Backbone mean differences decomposed with the fixed weights 0.1/0.3/0.6, using all 50 pairs, ten Normal pairs, and ten Hard pairs. The component values come from each run's final evaluation; their weighted sum gives the published composite score to the reported precision.

than Brief-errors on average. The figure shows all 12 observed batches, retaining their variation and original outcomes. These measurements include the batch's research lifecycle, waiting for calculations, and continuation after results return.

P2 effort is summarized over the 15 deeply inspected tasks shared by all four configurations. Guided used 11.08 job-hours per task, Standard 7.55, Record 7.88, and Brief-errors 15.59. The corresponding final composite means were 0.8124, 0.6495, 0.7360, and 0.7273. Guided used 46.7% more cumulative job time than Standard while increasing the mean score by 0.1629; relative to Brief-errors, it used 28.9% less job time and scored 0.0851 higher. Guided was both faster in cumulative job time and higher-scoring than Brief-errors on seven of the 15 paired tasks.

Across the common tasks, Guided submitted 789 recorded quantum-chemical jobs, Standard 419, Record 431, and Brief-errors 883. Recorded failed-job counts were 114, 45, 62, and 167, respectively, with cancellations recorded separately. The four difficult studies NU3, NU5, NU6, and NR1 accounted for 163.5 job-hours across their 16 runs, 24.3% of the 673.82 job-hours in the available 63-run effort archive. Some required steps in these four studies remained unresolved after additional attempts. In the archived Standard missing-graph cases, calculations could also outlive the research session, leaving subsequent results outside an active study.

Guided exceeded Brief-errors by 0.0268 overall, with an interval of $[-0.0486, 0.0996]$, 26 wins, eleven losses, and 13 ties. The comparison varied by research demand: Guided's mean advantage was 0.1514 on Normal and 0.1295 on Hard, whereas Brief-errors had the higher Impossible mean.

3.2.6 Uncapped electronic-barrier errors in a common numerical subset

To express numerical deviations in chemical units, we recovered electronic barriers directly from archived native TS and reactant-reference energies. A common subset contains 26 reference edges from 23 positive tasks with recoverable values in all four configurations and components matching the final publication records. Edge assignments follow the task's specified reaction and energy-reference definitions. Table 1 reports errors against the frozen task reference values, using each edge once per configuration and applying the energy conversion in Section 2.

Guided has a lower MAE and RMSE than Standard in this subset, while Brief-errors has the lowest MAE and RMSE. SI Section S4 gives every included value, the coverage of the 53 positive numerical reference edges, and a smaller subset with additional Gold-input method and medium corroboration. The numerical ordering varies with that coverage.

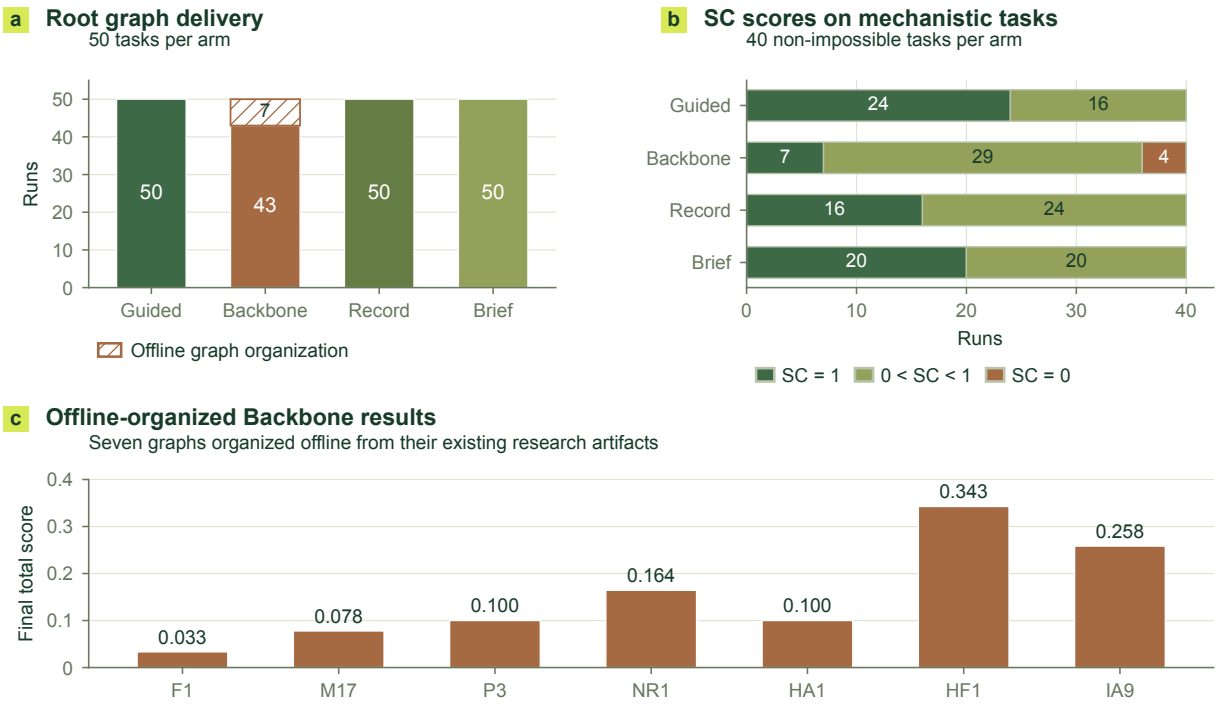


Figure 9. Runtime graph delivery and final scientific support in P2. (a) Original root mechanism graphs across 50 tasks per arm: Guided 50, Backbone 43, Record 50, and Brief 50. The hatched segment marks seven additional Backbone graphs assembled offline from the runs' own research materials; the organized set contains 50 graphs per arm. (b) Final continuous self-consistency (SC) scores for all 40 non-Impossible tasks per arm, separating SC = 1, partial positive credit, and SC = 0. Full-score counts are 24, 7, 16, and 20; Backbone has four zero-score tasks. (c) Organized composite scores for the seven Backbone tasks whose root graphs were reconstructed offline, ranging from 0.0333 to 0.3425 (mean 0.1537).

Table 1. Uncapped electronic-barrier errors for the same 26 reference edges in all four configurations, in kcal/mol.

Workflow	MAE	RMSE	Median $\lvert\Delta\rvert$
Guided	1.17	2.86	0.057
Standard	1.54	4.64	0.057
Record	1.58	5.71	0.073
Brief-errors	1.13	2.31	0.099

3.2.7 Per-task comparisons of barriers, pathways, and calculation evidence

Figure 11 places all 50 tasks and four configurations alongside their complete score components. NB1 and NI1 also illustrate the scores obtained on multistep mechanisms. NB1 scored 0.9628 under Guided, 0.9128 under Standard, and 0.9900 under Brief-errors; Guided and Brief-errors both had self-consistency 1. NI1 scored 0.9958 under Guided and 0.6196 under Standard, with self-consistency 1 and 0.35. In contrast, NQ5 favored Record over Guided, and NU3 favored Brief-errors.

The archived records explain several barrier differences (Figure 12). In HA8, Guided reported 29.46 kcal/mol and Standard 20.43 kcal/mol against a reference of 29.48 kcal/mol. The archive traces this 9.03 kcal/mol difference to the energy reference: the lowest all-s-trans reactant minimum in Guided and a reactive cisoid conformer in Standard. The IRC identifies the local structures connected by the saddle; the reported activation energy uses the stable-minimum reference specified by the task.

NU6 exposes a related issue in path identity. Guided's recorded concerted saddle had an energy of −247.030042392 Ha and the betaine minimum −247.060060699 Ha, giving a difference of 18.836772 kcal/mol. Guided described this as an energy cost involving the concerted saddle region while retaining the target betaine-to-product ring-closure saddle as unresolved. Standard delivered 18.837 kcal/mol as the second-step ring-closure barrier. The two reports assign this nearly identical energy difference to different processes. Guided's attempts from interpolated betaine structures and constrained scan peaks repeatedly relaxed toward the direct reactant-to-product saddle region. The saddle's path identity therefore required another assessment from its optimized geometry and pathway.

In G2, Guided scored 0.2389 and Standard 0.7769. Guided's native logs supported an approximately 0.845-kcal/mol double-H-transfer barrier. After three rejections of numerical barrier declarations on a companion edge, the agent removed the numerical fields. Supervisor feedback prompted restoration; a fourth rejection and a subsequent saddle-evidence check preceded their final removal, together with the saddle energy/log bindings. The final graph retained the values in text and received accuracy 0. Brief-errors retained the corresponding numerical result and scored 0.8500.

The accuracy values of 0.7 in D1 and D3 followed the specified cap on numerical-deviation penalties. D1's barrier difference from Standard traced to the reactant-conformer reference; D3's mainly reflected different saddle energies. The Supplementary Information provides the workflow-level components and native calculations for these case studies.

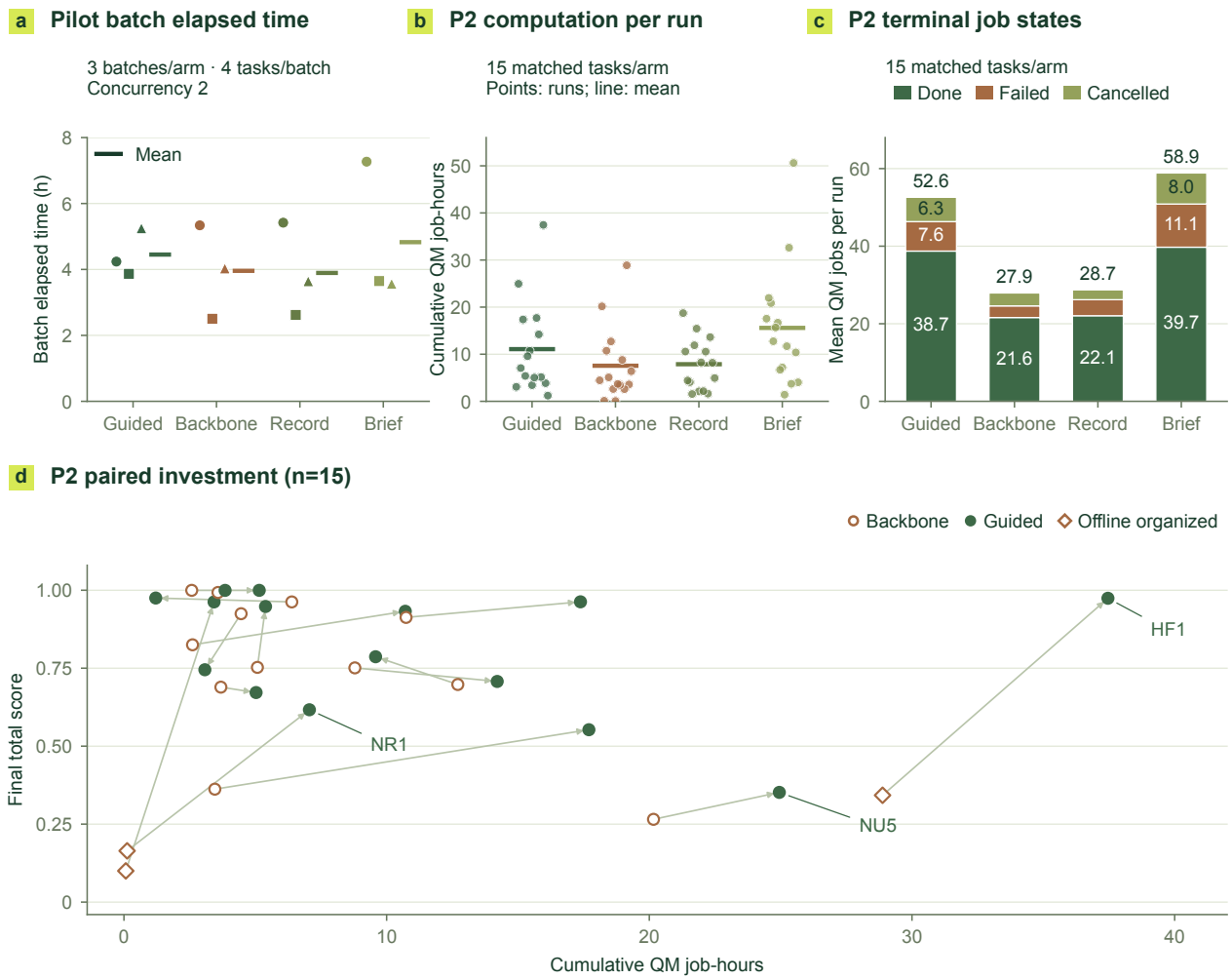


Figure 10. Research elapsed time and computational investment. (a) Twelve P1b four-task batches at concurrency two, three per arm; lines mark means. (b) Cumulative QM process durations for the 15 P2 tasks shared by all arms. (c) Mean completed, failed, and cancelled jobs in that cohort. (d) Same-task Backbone-to-Guided changes in investment and final score; diamonds mark offline-organized Backbone graphs. Job-hours sum process walltime; panel a measures batch latency. Backbone/M1 effort is reconstructed from native CPU time divided by four.

3.2.8 Impossible tasks distinguish local support from the requested negative conclusion

The Impossible tier tests whether research identifies the requested target as physically inaccessible under the permitted xTB/PM7 methods. Brief-errors averaged 0.6029, Guided 0.5051, Standard 0.2473, and Record 0.2925 (Figure 13). Brief-errors exceeded Guided on four tasks and scored lower on five, with one tie. The Brief-errors-minus-Guided mean was +0.09788, with a 95% task-bootstrap interval of $[-0.160, +0.361]$. Its weighted contributions were -0.00661 from graph, -0.01550 from self-consistency, and $+0.12000$ from accuracy. Task-level Brief-errors-minus-Guided differences were IA1 $+0.8283$, IA9 $+0.5666$, IC3 $+0.6000$, IF3 $+0.0333$, IA7 0, IA5 -0.0333, IB9 -0.0500, ID2 -0.1500, IE2 -0.1500, and IG8 -0.6661.

In IA1, concerning cyclobutadiene's square-geometry automerization target, Brief-errors had graph, self-consistency, and accuracy scores of 1, while Guided's accuracy was 0. In IA9, concerning dioxetane thermolysis, Brief-errors and Guided both had self-consistency 1, with accuracy 1 and 0, respectively. IG8 showed the opposite accuracy ordering for the chromium-dimer target. After target correspondence, Guided, Record, and Brief-errors received full negative-target Accuracy credit on IA5 (thermal disrotatory ring opening), and Brief-errors received full credit on IC3 (open-shell singlet phenylnitrene); Guided/IC3 retained Accuracy 0 with Graph and self-consistency both 1. Across the six tasks with the same accuracy score in Brief-errors and Guided, Brief-errors averaged 0.0583 lower. The higher Brief-errors mean therefore arose from the requested conclusions on specific tasks, while Guided retained stronger local support on several others.

Across all 40 Impossible runs, eight had accuracy 1 and ten combined self-consistency 1 with accuracy 0. IF3 illustrates the latter pattern: Brief-errors and Guided both had Accuracy 0 with self-consistency 1. Guided and Brief-errors both received partial accuracy of 0.7 in IB9, whose requested network contains positive and negative components.

3.3 Molecular identity, distance comparison, and fragment rotation

We next examine whether the model distinguishes molecular identities, compares distances between specified atoms, detects mismatched calculation references, and identifies missing IRC calculations. Archived research trajectories supply

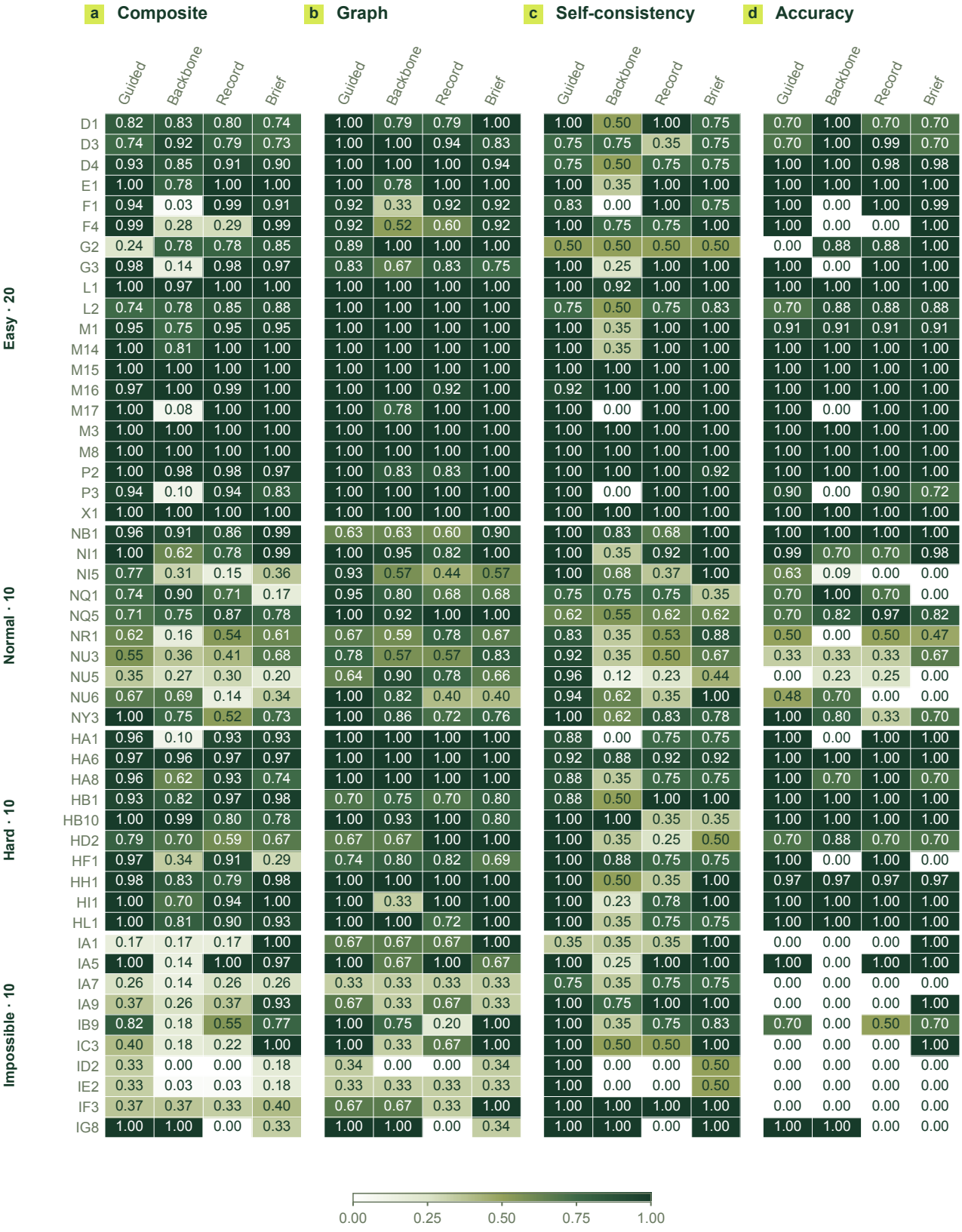


Figure 11. Complete CloudraBench results for all 50 tasks and four configurations. The aligned blocks show composite, Graph, Self-consistency, and Accuracy on the same 0–1 scale; all 800 cells are present. Values are rounded to two decimals; the source table retains full precision.

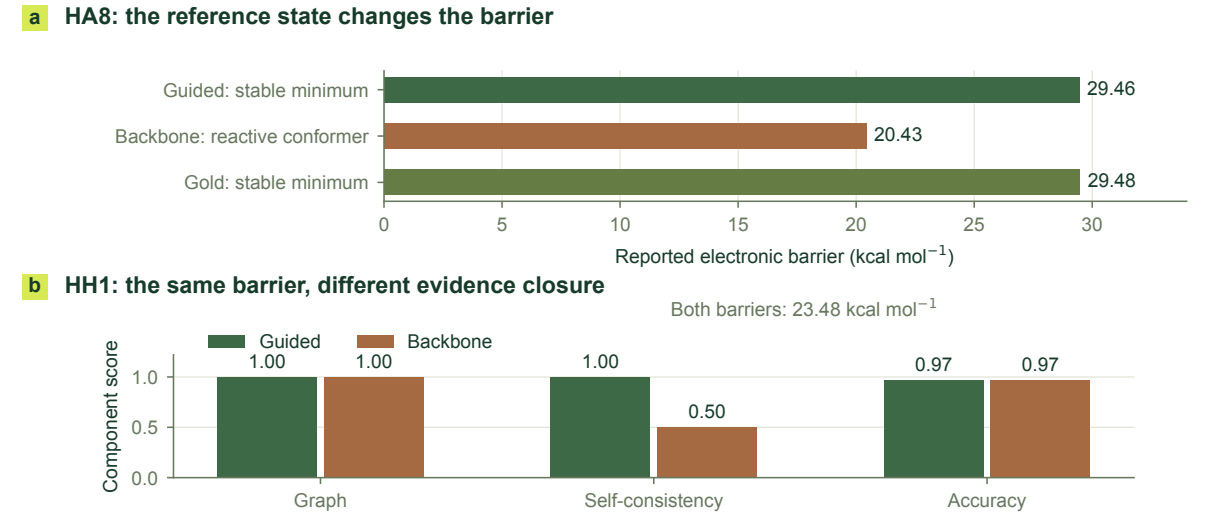


Figure 12. Energy references and pathway assignments in matched Guided–Backbone tasks. (a) HA8 barriers use a stable-minimum or reactive-conformer reference and are compared with campaign Gold. (b) Both HH1 runs report 23.48 kcal/mol with similar accuracy but different self-consistency scores. (c) NU6's recorded energy difference is 18.836772 kcal/mol: Guided reports the cost of reaching the concerted saddle region and leaves the target ring-closure saddle unresolved; Backbone reports 18.837 as the second-step ring-closure barrier.

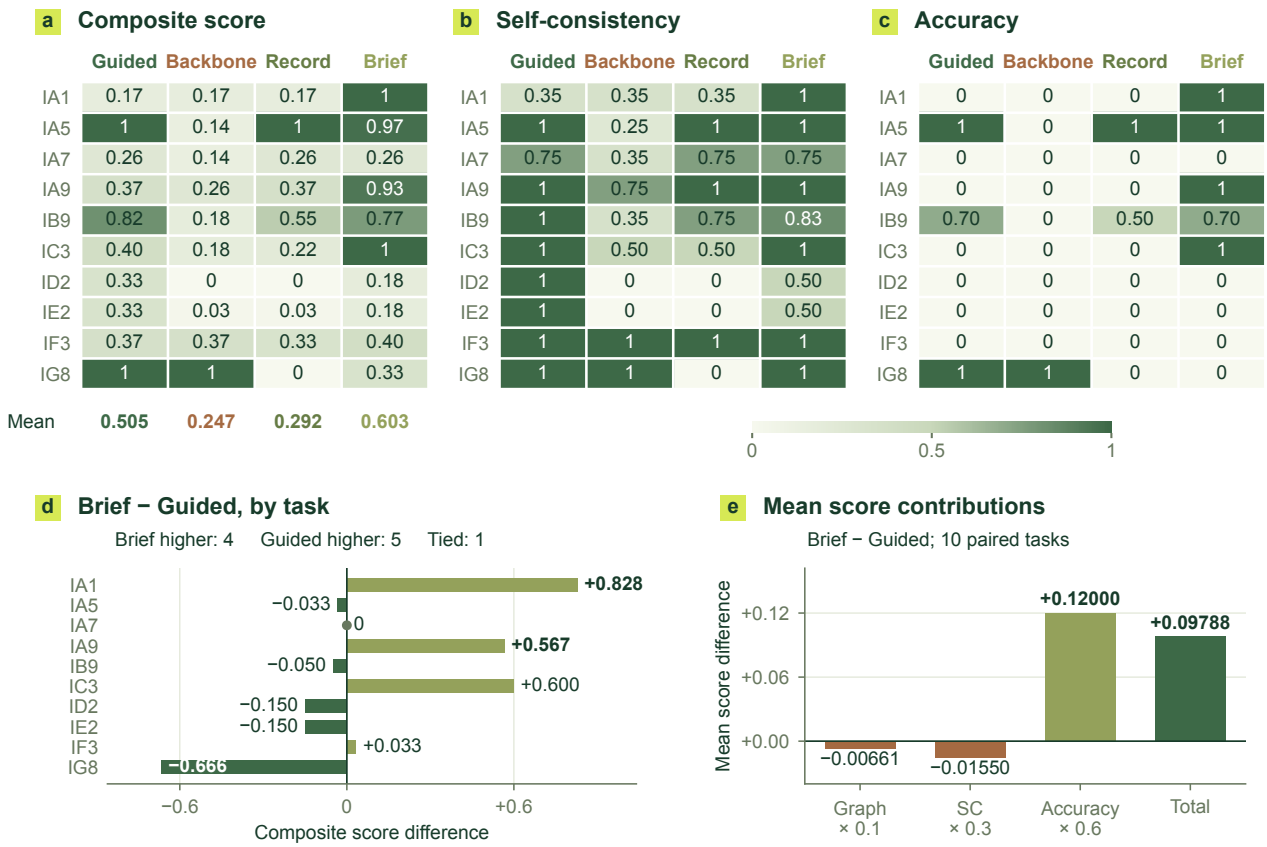


Figure 13. Impossible-tier outcomes after offline organization and scoring. (a–c) Composite score, self-consistency (SC), and accuracy for ten tasks and four arms, using a shared 0–1 color scale. Backbone denotes Standard. Brief averages 0.603 and Guided 0.505. (d) Guided scores higher on 5 tasks, Brief on 4, and 1 tie. (e) The +0.09788 mean difference comprises accuracy +0.12000, Graph −0.00661, and SC −0.01550 after weighting. The paired task-bootstrap 95% interval is −0.160 to +0.361.

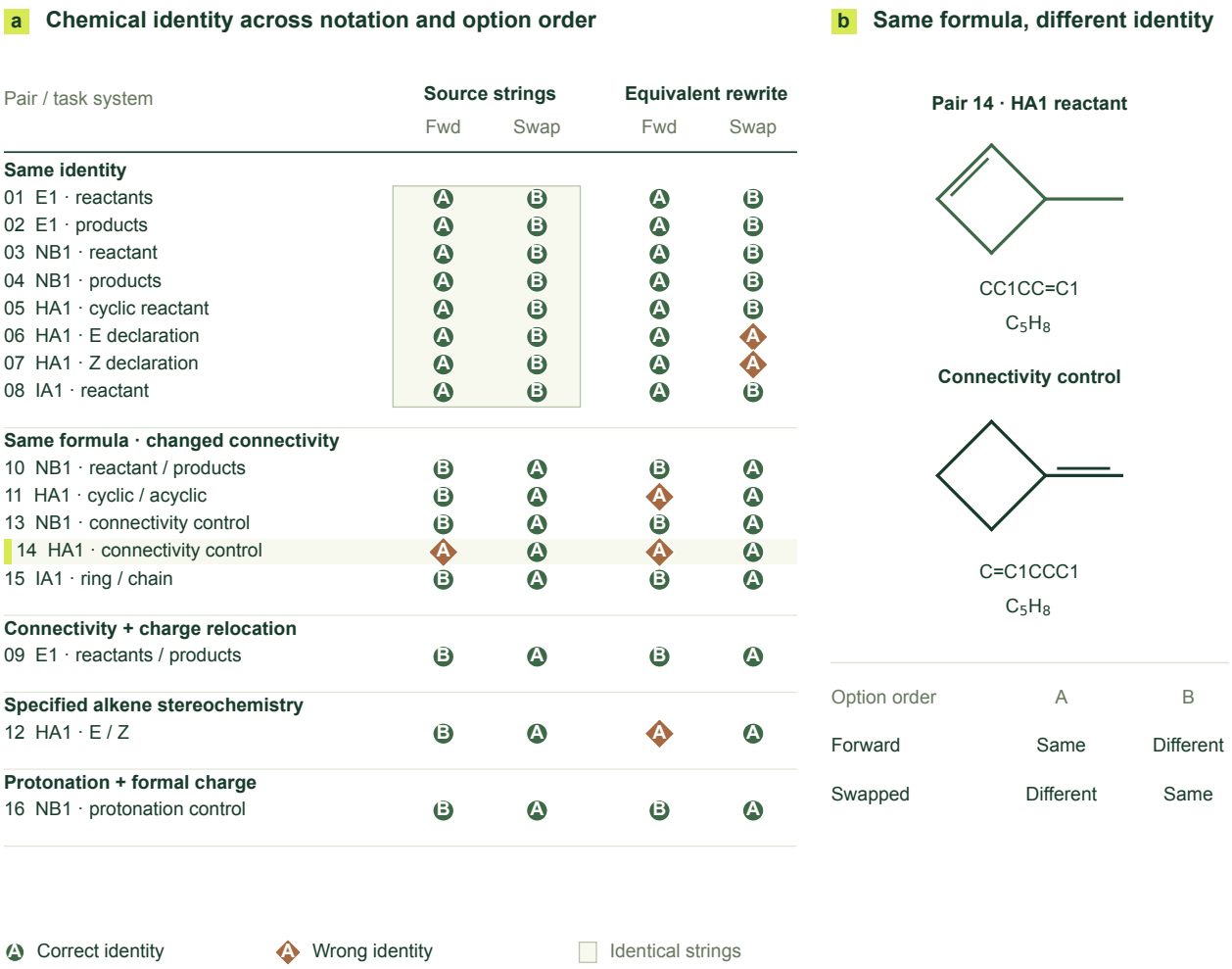


Figure 14. Molecular-identity judgments under changes in molecular representation and option order. Each of the 16 molecular pairs has four input conditions. Circles and diamonds indicate correct and incorrect judgments, respectively; the letters retain the model's actual answers. Shaded regions denote identical-string conditions. The right panel shows the pair-14 connectivities `CC1CC=C1` and `C=C1CCC1`, both C_5H_8, and the meanings of the answer options.

the molecular structures and calculation records. Controlled local studies with Qwen3.8:27b vary the supplied information, answer order, and selected internal states to test their effects on each judgment.

3.3.1 Equivalent SMILES and answer order affect molecular-identity judgments

A geometric modification first requires identifying the molecule, atoms, and reaction center to which it applies. A molecule can be represented by different SMILES traversals, while a molecular formula can describe different connectivities. To examine how the model handles these distinctions, we constructed 16 molecular-identity pairs covering identical identities, connectivity changes within the same molecular formula, changes in connectivity and charge location, double-bond stereochemistry, and protonation changes. Each pair was presented using the original strings or equivalent rewritings and with two answer orders, yielding 64 judgments.

The model answered 58 correctly. Equivalent rewriting changed the semantic judgment in 4 of 32 matched representation comparisons; reversing answer order changed the judgment in 6 of 32 matched option-order comparisons. All six of the latter cases involved repeated selection of the letter A. One C_5H_8 pair in Figure 14 illustrates this behavior directly: an endocyclic and an exocyclic double bond define different connectivities, yet the model answered A under all four conditions, although A meant "same" in one order and "different" in the other.

Cloudra stores molecular connectivity separately from coordinates and uses atom correspondence to identify which atoms to measure, move, or compare. Product-endpoint alignment, path interpolation, and reaction-center measurements retain this correspondence as the structure's coordinates change.

3.3.2 Geometric measurements reduce repeated numerical reasoning

We selected four actual structures from the NB1 trajectories and asked the model which of two specified interatomic distances was shorter. Each structure was presented with both answer orders and with either a complete XYZ geometry or program-measured distances. These two information conditions were further crossed with direct-answer and thinking modes, producing 32 native chat calls (Figure 15).

In direct-answer mode, all eight responses to XYZ inputs selected A, with four correct answers. Providing measured distances produced eight correct answers and preserved the

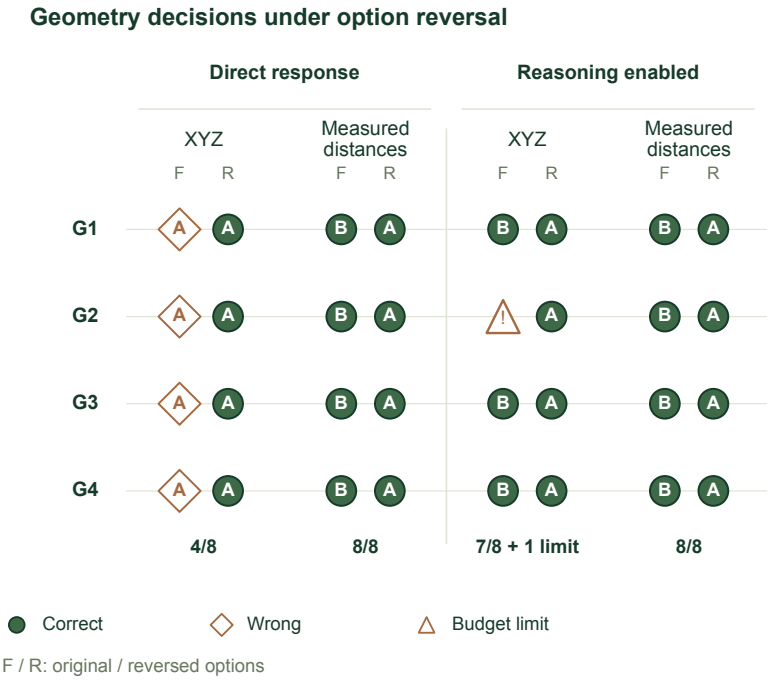


Figure 15. Judgments of which of two specified atom-pair distances is shorter, using coordinates or measured distances. Four source structures are compared under direct response and reasoning-enabled conditions, each with original (F) and reversed (R) option order. Circles indicate correct judgments, diamonds incorrect judgments, and the triangle a response that reached the output budget. Letters retain the actual answers.

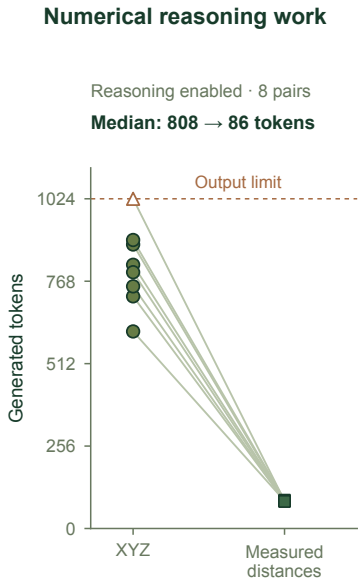


Figure 16. Generated-token counts with reasoning enabled. Each line connects Cartesian-coordinate input and measured-distance input for the same source structure and option order; eight paired comparisons are shown. The dashed line marks the output limit.

semantic answer when option order was reversed. With thinking enabled, the XYZ condition produced seven correct answers; the remaining response reached the output limit of 1,024 generated tokens. Increasing the budget for that input alone yielded a correct answer within 1,353 generated tokens.

All eight answers in the measured-distance condition were correct. Across the eight paired comparisons with thinking enabled, the median generated token counts were 807.5 for XYZ inputs and 86 for measured distances.

Providing the distances removes the need for the model to calculate them from XYZ coordinates. Cloudra similarly computes bond lengths, bond angles, and dihedral angles in software and uses deterministic operators to rotate or translate the selected fragments. The model chooses the atoms and target geometry, then compares the measured distances or angles of the resulting candidates. Assigning these numerical operations to software follows the approach of program-aided language models.[40]

3.3.3 Feedback must match the specified atoms and geometric objective

In an actual E1 trajectory, the model sought to bring the dihedral angle defined by the transferring hydrogen H7, C2, C3, and leaving atom Cl4 close to 180°. It rotated the fragment on the C3 side about the C2–C3 axis by +120° and −120°, producing candidates A and B, and set the C3–Cl4 distance to 2.35 Å. The representative dihedral in the default structure report involved H5, whereas the current objective involved H7. The model then explicitly measured H7–C2–C3–Cl4 in both candidates, obtaining approximately 59.4° and 179.4°, and used candidate B in its next actual submission.

Figure 17 shows Newman projections along C2→C3 drawn directly from the saved candidate coordinates, and Figure 18 summarizes the corresponding target-angle residuals. Candidate B places the H7 target angle within 0.6° of 180°, compared with a residual of 120.6° for A.

The corresponding residuals for the H5 comparison were 124.2° and 115.8° in A and B, respectively. Identifying the anti-periplanar H7–Cl4 arrangement in B required the H7-based dihedral.

When local probes provided only H5 measurements, the model requested an additional H7 measurement under all three cyclic option permutations. Adding H7 measurements led it to select B under all three permutations. Two further paired tests of information sufficiency also shifted from requesting additional measurements to judging the information sufficient when the target quantity was added.

3.3.4 Activation replacement affects distance answers differently across option orders

To test how a reported distance affects the answer, we performed activation replacement at a single token position in the same local quantized model.[41] One distance in the input was changed from 2.029 Å to 1.029 Å, while the other remained at 1.248 Å, reversing the correct comparison answer. We then transferred the residual-stream state at the numeric token position from the original input into the modified input at layers 7, 15, 31, and 47, and read the full-vocabulary probability assigned to the answer supported by the original fact.

The effect varied with layer and option order (Figure 21). With reversed options, the probability of A, the original-fact answer, increased from 0.074 for the modified input to 0.956 and 0.718 after replacement at layers 7 and 15, respectively. Replacements at layers 31 and 47 yielded 0.072 and 0.070. With the original option order, the probability of answer B changed from 0.009 to 0.029 and 0.015 after replacement at

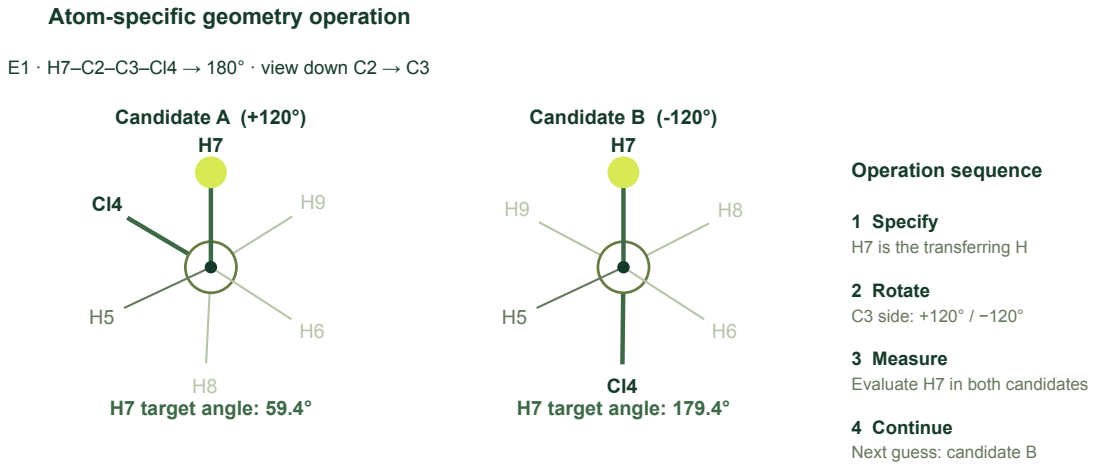


Figure 17. Target-specific geometry operations in E1. Newman projections retain the directions of the original candidate coordinates viewed down C2→C3. H7 is the specified transferring atom. Candidate A and candidate B give target angles of 59.4° and 179.4°, respectively. The operation sequence is shown alongside the projections.

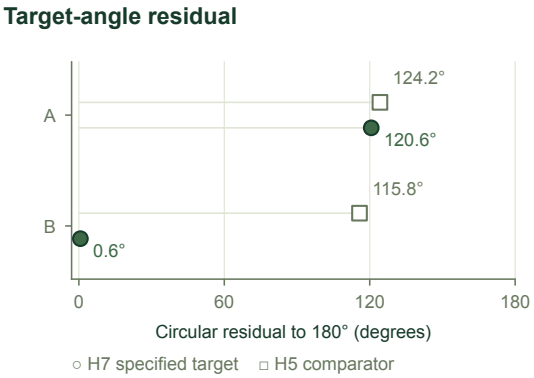


Figure 18. Circular angular residuals from the specified 180° target for candidates A and B. H7 is the specified target atom; H5 is the comparator.



Figure 19. Next-action choices under three local option permutations. When only H5 measurements are supplied, the model requests H7 measurements; with H7 supplied, it selects candidate B.

the first two tested layers, while the last two remained close to the modified-input baseline.

Self-replacement controls and control replacements at an unchanged prefix position produced zero logit differences across the full vocabulary.

With reversed options, replacing the early-layer state restored the original-fact answer as the preferred choice; with the original option order, the modified-fact answer remained preferred. Recovering the answer from the same numeric position therefore depended on option order, as did the identity judgments in Section 3.3.1.

3.4 Identifying the species and pathways established by calculations

3.4.1 Associating frequencies and IRC endpoints with the same saddle

The frequency calculation and IRC must use the declared saddle, and each endpoint optimization must start from the corresponding IRC terminal frame, followed by frequency analysis of the resulting minimum. The DAG links these calculations to the species and step and records their execution, convergence and physical checks.

We extracted four source evidence chains from the archive and constructed versions that preserved the links or changed a single reference. Crossing these with flat and linked representations and two answer orders yielded 32 inputs. The altered references concerned the frequency geometry, IRC starting geometry, endpoint-optimization input, or target species; local termination, convergence, energy, and frequency values were held constant. Figure 23 illustrates the E1 case: changing the frequency geometry reference from the saddle point to an existing endpoint changes how that frequency result supports the current step.

All 16 judgments with linked representations were correct, compared with 14 of 16 with flat representations. Both errors occurred when a reference had been changed and the options reversed, in E1 and the second NB1 step. In subsequent native follow-up calls on two selected source chains, enabling thinking yielded correct judgments for all 16 judgments of the evidence relationships, including the two previously incorrect cases.

The model detected every altered reference in the linked format, even though the local numerical values were unchanged. This presentation effect is consistent with observations that graph encoding affects relational reasoning in language models.[42] Cloudra links each species, step and calculation attempt to its calculation results, showing which saddle, IRC and endpoint calculations belong together.

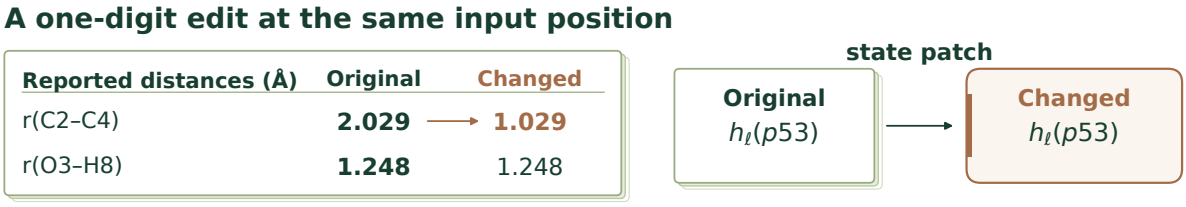


Figure 20. Intervention on a single reported distance. A one-digit edit changes the C2–C4 distance from 2.029 to 1.029 Å while O3–H8 remains 1.248 Å. The original-input residual state at token position 53 is inserted into the changed-input run at a specified layer. Token positions are zero-based.

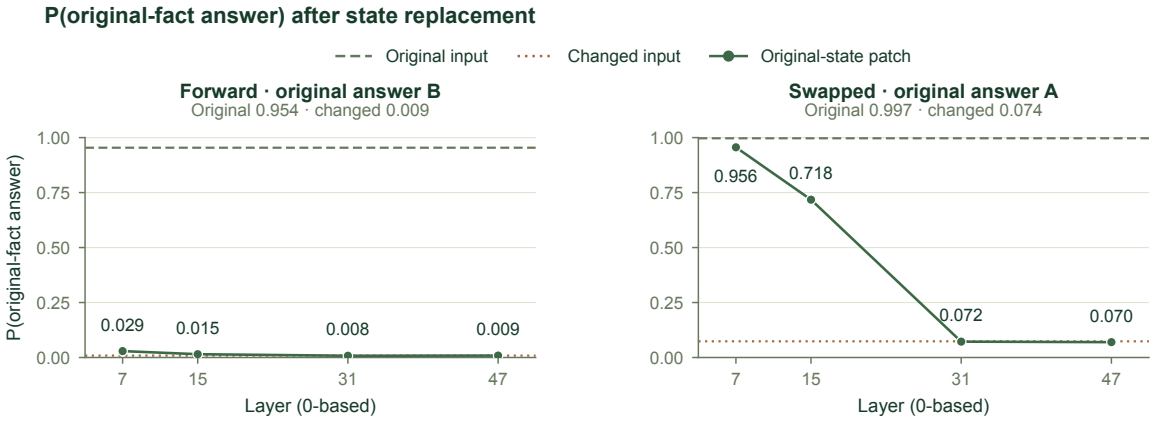


Figure 21. Original-fact answer probability after state replacement. Curves show full-vocabulary probability mass for the original-fact answer under forward (B) and swapped (A) option order, including bare and leading-space token spellings. Horizontal lines show original-input and changed-input baselines. Layer indices are zero-based.

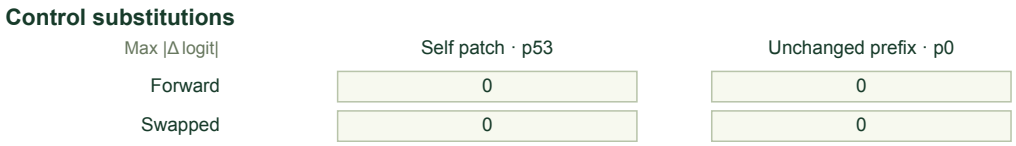


Figure 22. Control substitutions for the state-intervention experiment. Same-state self replacement at p53 and replacement of the unchanged prefix at p0 preserve all vocabulary logits in both option orders. The table reports the maximum absolute logit difference. Token positions are zero-based.

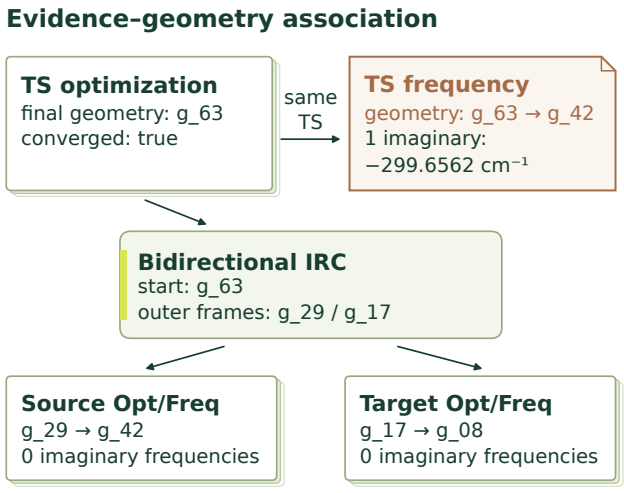


Figure 23. Changing the geometry assigned to the E1 frequency calculation. The frequency geometry reference changes from g_63 to g_42 while local termination, convergence, energy, and frequency values remain fixed. Arrows connect the TS, IRC, and endpoint optimization/frequency calculations.

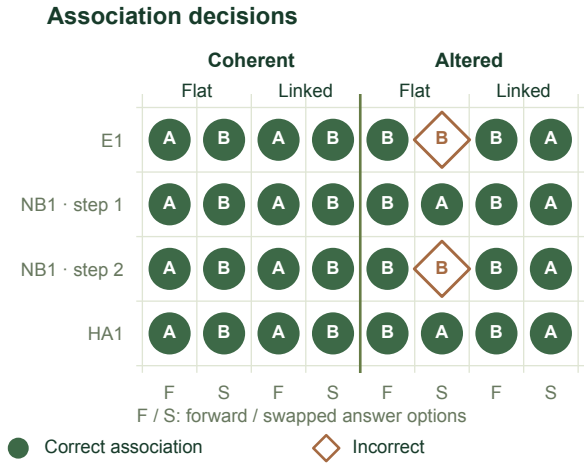


Figure 24. Judgments of whether calculation references support the specified step. The 32 inputs cross four source chains, coherent/altered links, flat/linked presentation, and forward/swapped answer order. Saved A/B responses and correctness markers are retained.

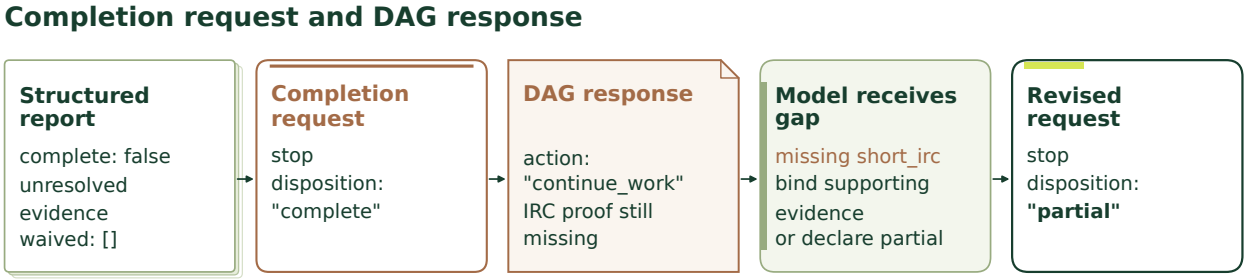


Figure 25. Completion requests and DAG feedback in the original IA1 trajectory. A report with unresolved evidence is followed by a completion request, a response identifying missing short_irc proof, feedback to the model, and a revised partial-completion request.

3.4.2 Diagnostic feedback on IRC connectivity and transition-state curvature

Diagnostic feedback connects an evidence defect to the chemical object and calculation that require revision. Cloudra's structured diagnostics identify specific problems in quantum-chemical outputs, including steric-clash atom pairs, imaginary-mode misalignment, and disconnected trajectory endpoints. The agent can then change the clashing geometry, inspect the imaginary mode, or recalculate the affected endpoint.

Two archived trajectories illustrate this connection. In a Brief-errors HA1 trial, compact error codes replaced the detailed IRC-endpoint linkage diagnostics. The Computational Agent repeatedly substituted source-minimum references while the geometric disconnection remained unresolved. In a Guided NB1 trial, the verification found zero imaginary frequencies and returned a normal-mode diagnostic. The Coordinator requested a retry with revised transition-state guess parameters and optimization constraints, which the Computational Agent used in the rescheduled calculation.[43]

3.4.3 IA1: a completion request with a missing IRC calculation

An archived IA1 trajectory shows how a completion request can conflict with the recorded scientific requirements. The reaction step had mandatory bidirectional short_irc verification pending, and the waiver list was empty. The Supervisor had earlier commented positively on intermediate optimization convergence. The model subsequently requested completion while the IRC requirement remained unresolved.

The harness rejected that request and returned the outstanding short_irc requirement to the model. In the next submission, the model changed its status to partial completion (Figure 25).

We also tested whether the model could identify the outstanding IA1 requirements from the recorded state. All 12 judgments were correct (100%) across six pairs with reversed answer-option order, using both a minimal summary and an extended 12-message history.

3.5 Workflow guidance, scientific judgment, and outlook

Across the 50-task comparison, Guided had higher mean self-consistency and accuracy than Standard. On Hard tasks, their weighted contributions to the gain were +0.1353 and +0.1274. The HH1 and HI1 results show how IRC and endpoint calculations establish the connections behind reference-consistent barriers.

HA8 and NU6 show why energy references and saddle assignments must follow the computed structures and paths. In HA8, the stable reactant minimum defines the barrier reference, while the IRC connects the saddle to a local reactive conformer. In NU6, guesses for betaine-to-product closure repeatedly optimized toward the direct reactant-to-product saddle region. The DAG keeps the reactant conformer, saddle, IRC endpoints and assigned step together so that the agent can check which structure each energy belongs to after a new calculation. The calculations provide external feedback for the agent's planning and verification.[44]

These cases call for different next steps: calculate a missing endpoint, revise a guess that returns to another route, or reassess a method that cannot describe the requested electronic state. The agent must update its reaction description when the computed structures or paths change.[45] Record retained the tools and obtained lower self-consistency than Guided, particularly on Normal and Hard tasks. The archived trajectories show how tool feedback directs the next geometric operation or calculation.[43, 46]

Brief-errors scored higher on IA1 and IA9, whereas Guided scored higher on IG8. These crossed outcomes motivate a task-dependent hypothesis: detailed diagnostics may help repair a TS or IRC calculation, with limited benefit from further local repair when the method or target needs reconsideration.

In G2, repeated checks led the agent to delete numerical values already supported by native calculations. Supervisor feedback restored the fields before further rejection led to their final deletion. A numerical observation and its log should remain attached to the graph while its mechanistic assignment is being tested.

P2 contains one run per task and configuration; its task-cluster bootstrap intervals describe variation across tasks. P1b provides three repeats on each of four tasks under each configuration, totalling 48 runs. Some Hard-task Accuracy targets score qualitative channel identity, so the reported gain includes pathway identification and evidence completion as well as numerical barriers.

Standard sometimes ended research while calculations were still running, whereas Guided/HI1 remained waiting after the completion notification. Returning results therefore need to resume the unfinished study before final reporting. NU6's repeated convergence to one saddle region also motivates comparing new candidates with previous attempts before choosing another guess.

Default preorganization samples 32 rigid relative placements and selects locally relaxed MMFF94 candidates by

energy and clustering. CREST provides additional sampling through an optional backend.

The benchmark scores electronic energy differences, including the prescribed references for NI5 in weakly acidic aqueous solution and NU5 in concentrated dichloromethane. Solution-phase $\Delta G^{\ddagger}$ profiles would include zero-point, thermal/entropic, standard-state and solvation contributions. Solvent participation can also change the chemical model used to describe a pathway; solvent sensitivity remains to be assessed.

The workflow comparisons use fixed atom-mapping and branch-crowding settings and the DeepSeek-v4.1-flash (1017) backbone, with API alias `deepseek-v4-flash`.

Cloudra is publicly available as open-source software under the GNU Affero General Public License (AGPL) at `https://github.com/yyywaa/Cloudra-mechanism-agent`, supporting inspection, reuse, and community development of the system. Reproducible tables and scores are in the companion data package.

Data availability

The retained P2 benchmark records, the 48-run P1b study, Gold reference-calculation evidence, and supplementary reproduction materials are publicly available at `https://drive.google.com/drive/folders/1yZo3eJuCUuAgedQztgCtLyJcc8MZipf7`. Package contents, versions, and checksums are described in Supplementary Information Section S7 and the accompanying download records. The source code is publicly available at `https://github.com/yyywaa/Cloudra-mechanism-agent` under the AGPL; revision 70daa5a identifies the described implementation. Reproducible tables and scores are in this companion data package.

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